

When policy changes the channel

Three rigorous economics papers · two current applied-development studies and one foreign-aid classic. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

These papers share a deceptively simple message: policy inputs do not map mechanically into outcomes because social institutions determine the channel. Mali's HIV law appears to have amplified stigma and reduced women's testing. Electrification in rural Iran raised fertility where women had few market opportunities, reversing the usual income–substitution balance. The foreign-aid classic shows how evaluation horizons can similarly reverse a verdict when human-capital returns arrive slowly. **Literature trend:** applied economics is moving beyond average treatment effects toward context, mechanisms, and timing. A date-based regression discontinuity, an elevation instrument, and a calibrated growth model illuminate different margins, but each conclusion remains tied to a specific identifying assumption: no coincident cutoff shock, no direct fertility effect of relative elevation, and credible macro parameters.

01 CURRENT / GLOBAL HEALTH AND LAW · STARTS ON PAGE 2 · *Journal of Demographic Economics* · 2026 · peer reviewed · CC BY 4.0

Impact of Legislation for Infectious Disease Control: Evidence from HIV Testing in Mali

Can a public-health law discourage the behavior it is meant to promote? Mali enacted an HIV-specific law during the 2006 Demographic and Health Survey, which offered voluntary testing. A regression discontinuity in interview dates finds that women's testing fell by about 3–4 percentage points at enactment, with larger declines in rural areas and among women with radios or little schooling; men's testing did not change. The paper interprets the pattern as an expressive effect of criminalization that intensified stigma. The estimate is local to one legal cutoff and a discrete survey date, so coincident changes remain the key caveat.

02 CURRENT / INFRASTRUCTURE AND DEMOGRAPHY · STARTS ON PAGE 17 · *Journal of Demographic Economics* 92(3) · 2026 · peer reviewed · CC BY 4.0

Rural Electrification and Fertility Decline in Iran

Does electricity always accelerate fertility decline? Using roughly 45,000 Iranian villages observed in the 1986, 1996, and 2006 censuses, the authors instrument years of electrification with a village's elevation relative to its district. OLS suggests lower fertility, but the IV estimates reverse sign: each year of exposure raises the child–woman ratio by about 22 per 1,000 in 1996 and 16 per 1,000 in 2006, with larger effects where women's labor-force participation is low. The child–woman ratio is an imperfect fertility measure and the elevation exclusion restriction is contestable, but the result is a sharp reminder that infrastructure effects depend on women's economic opportunities.

03 CLASSIC / FOREIGN AID AND GROWTH · STARTS ON PAGE 32 · *World Bank Economic Review* 30(3) · 2016 · peer reviewed classic · CC BY-NC 4.0

What Is the Aggregate Economic Rate of Return to Foreign Aid?

What return would foreign aid earn if evaluated like a long-lived public investment? The authors benchmark a dynamic growth model to the post-2008 aid–growth literature and run 5,000 simulations spanning physical capital, human capital, productivity, lags, and aid consumption. Short-horizon growth effects are noisy and can be negative, but 30-year effects are consistently positive; the mean internal rate of return is about 11.2% when all channels vary. This is disciplined calibration rather than a new causal estimate, so conclusions inherit the macro literature and parameter choices. Its enduring lesson is that judging aid over five years can badly understate slow human-capital returns.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

RESEARCH PAPER

Impact of legislation for infectious disease control: evidence from HIV testing in Mali

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Abstract

This study examines the impact of human immunodeficiency virus (HIV)-specific laws criminalizing HIV non-disclosure, exposure, and transmission on voluntary testing, focusing on the role of HIV stigma. HIV criminalization signals state endorsement of discrimination against HIV-positive individuals, thereby amplifying stigma. I use a regression discontinuity design that exploits the enactment timing of legislation in Mali during a household survey offering voluntary HIV testing, where family members could infer who was tested and speculate that those tested were HIV-positive. Following the legislation, women's testing uptake declined, especially in rural areas, with stronger effects among those with radios and without completed formal education. Women, being economically dependent on men, are vulnerable to HIV-related mistreatment from family members. Therefore, fear of being considered seropositive by family members might have more strongly discouraged women's testing uptake compared with men's.

Keywords: Discrimination; HIV criminalization; infectious diseases; stigma; voluntary testing

JEL classification: I12; I18; J16; K32

1. Introduction

History has seen major infectious diseases such as acquired immune deficiency syndrome (AIDS), cholera, Ebola, leprosy, tuberculosis, and COVID-19. The economic impact of these diseases (Malani & Laxminarayan, 2011) has spurred debates on whether criminalizing behaviors that facilitate their spread is effective (Burris 2006; WHO, 2015). Such legislation often exacerbates discrimination against those infected, deterring serostatus disclosure, care-seeking, and self-protection, which increases the overall societal health burden (Chandrashekar, 2020; Valdiserri, 2002).¹ However, rigorous empirical studies on the impact of such legislation are limited.

¹This is known as the “expressive” effect of the law; it influences behaviors by shaping cultural values, even if enforcement (e.g., monitoring and legal punishment) is weak (Lane *et al.*, 2023; Sunstein 1996).

This study examines legislation in Mali aimed at reducing the spread of human immunodeficiency virus (HIV), focusing on its impact on voluntary testing through the underlying influence of HIV stigma. Recently, over half of sub-Saharan African countries have enacted similar laws (Eba, 2015). These laws combine protective provisions for alienated groups, such as counseling for persons living with HIV (PLHs), men who have sex with men (MSM), and sex workers, with punitive provisions like criminalizing HIV non-disclosure, exposure, and transmission. Public health experts and advocates argue that such criminalization worsens discrimination against these already marginalized groups, discouraging serostatus disclosure and the practice of HIV-protective behaviors, such as voluntary testing (Bernard & Cameron, 2016; Eba & Lim, 2017).

To explore this issue theoretically, I develop a model in which two types of agents with different beliefs about their HIV status – based on past sexual behavior – decide whether to undergo HIV testing. In this community with a fixed fraction of HIV-positive individuals and existing discrimination, testing incurs two costs related to HIV stigma: reduced self-esteem and fear of peer discrimination if tested individuals are perceived as HIV-positive. This perceived cost is calculated as the exogenous incremental cost of being viewed as a PLH, multiplied by the peers' endogenous belief that tested individuals are HIV-positive. HIV criminalization increases these stigma costs by signaling state endorsement of discrimination. The model thus predicts that, all else being equal, HIV criminalization discourages testing, especially among those who more strongly care about the stigma costs (likely, HIV-positive individuals).

I assess the impact of the law using data from the 2006 Standard Demographic and Health Survey (DHS), which offered informed, anonymous, and voluntary HIV testing. This setting is ideal for three reasons.

First, Mali's HIV-specific law was enacted on June 29 (hereafter, referred to as the day of legislation) during the survey period (April to December 2006). There is no discontinuity in the number of respondents on that date, and covariate checks show similar characteristics for respondents interviewed just before and after the legislation. This permits the use of a regression discontinuity (RD) design with the interview day as the running variable. The temporal coincidence of the law's enactment and the survey provides a rare and powerful quasi-experimental setting for causal identification. In contrast, other common non-experimental methods, such as difference-in-differences, are less suitable because HIV-specific legislation usually applies nationwide – making it difficult to identify a valid control group – and because long-term panel data would be required to control for time trends. Finding a valid instrumental variable for the legislation would also be challenging. To the best of my knowledge, no other African country has conducted a Standard DHS with HIV testing modules concurrently with the enactment of an HIV-specific law, making Mali's 2006 survey uniquely informative.

Second, the DHS's testing aimed solely at HIV surveillance, with results not linked to individuals and no feedback given to tested individuals. This minimizes the impact of fear of legal punishment and the law's protective measures on testing behavior.

Third, the consent process often lacked privacy in practice, potentially allowing household members to infer who was tested and leading to speculation about the seropositivity of tested individuals and subsequent discrimination. Therefore, HIV stigma influenced respondents' decisions to accept or refuse testing.

Empirical analyses show a 3%–4% decrease in women's HIV testing uptake on the day of legislation compared to the previous day. The decline was greater in rural areas – where stigma might be more impactful due to limited job opportunities, stronger mutual

support networks, and greater visibility of testing status – compared to urban areas. Additionally, the effect was stronger among those with access to radios, suggesting that radio broadcasts heightened awareness of the new law. Furthermore, the legislation primarily discouraged testing among women without completed formal education, who may have misunderstood the testing procedures – particularly the anonymity and confidentiality of test results – and thus experienced stronger stigma-related concerns. In contrast, the law did not significantly affect men's testing uptake. These findings remain robust across various bandwidths and polynomial orders, as well as through standard robustness checks, including the donut-hole exercise and permutation test. Using alternative interview dates as hypothetical cutoffs did not produce similar results, and the decline in women's testing uptake is not significantly affected by specific refusals.

This gendered difference may be explained by high gender inequality and discrimination in Mali (OECD, 2023; Sida, 2004) and prior research. Malian women, with limited financial independence (Heath *et al.*, 2020; Lees *et al.*, 2021; Van de Walle, 2013), are vulnerable to HIV-related mistreatment from family members, increasing their incentive to avoid being perceived as a PLH. Thus, the law may have exacerbated this incentive, disproportionately deterring women from testing. This view supports existing claims that HIV criminalization disproportionately harms women (Ahmed, 2011; Global Commission on HIV and the Law, 2011; Weait, 2011), as women, frequently the first to discover their HIV infection through antenatal or other healthcare services, face blame for HIV transmission and are at higher risk of abuse from family members, leading to reduced willingness to seek healthcare services after the legislation.

This study contributes to the literature in four key ways. First, while existing research has explored the association of HIV-specific legislation with attitudes toward the law, disclosure, testing, safer sex practices, and stigma (for literature reviews, see Harsono *et al.*, 2017 and O'Byrne *et al.*, 2013), this study evaluates the causal impact of such legislation.² Unlike many studies focused on PLHs and at-risk groups in the US (e.g., drug users, heterosexuals, immigrants, MSM, and sex workers), it also examines representative populations in Africa, where HIV is a major issue, and offers a fresh perspective by analyzing actual testing decisions rather than self-reported experiences.³

Second, this study advances the literature on HIV-prevention programs, particularly those targeting HIV stigma to encourage voluntary testing and reduce risky behavior. Despite the importance of this issue, few studies have rigorously evaluated the impact of such programs on HIV stigma (Arimoto *et al.*, 2016; Godlonton & Thornton, 2012; Thornton, 2008; for literature reviews, see Andersson *et al.*, 2020 and Sengupta *et al.*, 2011). This study suggests that HIV stigma hinders testing uptake and that HIV-specific laws may have effects similar to some nonfinancial (see McCoy *et al.*, 2010 for literature reviews) and financial interventions (Kohler & Thornton, 2011) that have not significantly reduced HIV incidence.⁴

²For example, Lee (2015) found that media coverage of HIV criminalization was negatively associated with self-reported HIV testing experiences for at-risk individuals in the United States (US), but the existence of HIV-specific statutes was not.

³Furthermore, HIV-specific legislation, similar to HIV/AIDS information campaigns (de Walque, 2007), relies on public access to legal information and aims to influence behavior related to healthcare demand and supply (de Walque *et al.*, 2015).

⁴Some financial interventions have shown promise (Baird *et al.*, 2012; Björkman Nyqvist *et al.*, 2018; de Walque *et al.*, 2012).

Third, this study extends research on HIV stigma, including conceptual frameworks (Maughan-Brown, 2006; Parker & Aggleton, 2003), quantitative measures (Berger *et al.*, 2001; Hoffmann *et al.*, 2014; Kalichman *et al.*, 2005), and the stigma-testing relationship (Haffjee *et al.*, 2018; Kalichman & Simbayi, 2003; Maughan-Brown & Nyblade, 2014; Young & Bendavid, 2003). It specifically examines how law-induced stigma affects HIV testing uptake.

Fourth, this study highlights the unintended consequences of public policies, consistent with previous research (Cameron *et al.*, 2021; Doleac & Hansen, 2020). For example, Gertler and Shah (2011) found that licensing regulations in Ecuador's brothel sector led some sex workers to migrate to riskier street sectors, increasing sexually transmitted infection rates in both sectors.

The remainder of this paper is organized as follows. Section 2 describes Mali's HIV-specific law. Section 3 presents the conceptual framework for understanding how HIV-specific legislation influences voluntary testing and assesses whether HIV testing protocols in the DHS help address this study's question. Sections 4 and 5 detail the data and empirical strategy, with findings presented in Section 6. Section 7 concludes the paper.

2. Background

2.1. HIV-specific law in Mali

On June 29, 2006, Mali enacted its HIV-specific law (Loi n° 06-028; see Section S.6 of the supplementary material) upon the signing by President Amadou Toumani Touré. This law was based on the N'Djamena Model Law, developed and promoted by Action for West Africa Region-HIV/AIDS, a nongovernmental organization (NGO) funded by the United States Agency for International Development. The model law aimed to standardize legal frameworks and best practices for addressing sexually transmitted infections and HIV/AIDS in western and central Africa. Named after N'Djamena, Chad, where it was finalized in 2004 at a meeting with parliamentarians from 18 countries, the N'Djamena Model Law was intended to be adapted to each country's specific context. However, it was rapidly adopted by many countries (Grace, 2015), with most passing legislation closely mirroring the model law (Kazatchkine & Kra, 2020).

The HIV-specific law in Mali is omnibus legislation similar to the N'Djamena Model Law. It was originally designed to protect marginalized groups, including PLHs, MSM, and sex workers. Accordingly, it features protective measures on HIV-related education, equal access to healthcare and support, informed consent for HIV testing, confidentiality of test results, pre- and post-test counseling, and a prohibition on discrimination against PLHs.

However, the law also criminalizes HIV non-disclosure, exposure, and transmission. For instance, PLHs are required to disclose their serostatus to their spouse or sexual partners within 6 weeks of diagnosis (Article 27). Violations of this requirement can result in imprisonment for 6 months to 2 years and/or a fine of CFA francs 50,000 to 500,000 (Article 29).⁵ Additionally, the law imposes sentences of 5 to 20 years for PLHs who intentionally expose others to HIV (Article 37). This provision aims to deter intentional transmission, but proving intent is often difficult. The law's vagueness might also lead to criminalizing non-malicious exposure, including mother-to-child

⁵This amount is approximately USD 84 to 846 based on the exchange rate as of May 13, 2025. This fine is significant relative to Mali's gross national income per capita, which was USD 830 in 2020 (<https://data.worldbank.org/indicator/NY.GNP.PCAP.CD>).

transmission (Csete *et al.*, 2009). Furthermore, Articles 18 and 25 allow for compulsory HIV testing under certain conditions (e.g., for sexual offenders and pregnant women, or in marital disputes), involuntary notification of HIV status to partners by healthcare professionals, and potential discrimination based on HIV status (e.g., by partners or parents).

The protective provisions are intended to encourage voluntary testing among at-risk individuals, particularly those who are HIV-positive. However, the restrictive and punitive provisions discourage testing in three ways. First, at-risk individuals may avoid HIV testing to evade criminal liability for non-disclosure and intentional transmission, which they believe would apply only if they were aware of their serostatus. Second, the fear of involuntary notification of seropositivity to sexual partners, along with potential consequences like discrimination and legal prosecution, reduces testing uptake. Third, these provisions exacerbate HIV stigma by publicly endorsing discrimination against PLHs, which deters at-risk and HIV-positive individuals from engaging in protective behaviors, seeking appropriate healthcare, and disclosing their serostatus (Eba & Lim, 2017; Pearshouse, 2007). This study focuses on the third mechanism, as the test setting described in subsection 3.2 narrows the scope, and HIV stigma is widespread in Mali (Castle, 2003, 2004).

Finally, the law includes multiple provisions, each considered a distinct treatment. This study cannot identify which specific provision increases or decreases HIV stigma. Nonetheless, the overall impact of these provisions on HIV stigma and testing remains important from a policy perspective.

2.2. Public awareness

For the HIV-specific law to influence testing uptake, it must be assumed that citizens were aware of the legislation. However, DHS data do not include information on such awareness, and research on public knowledge of the legislation in Mali is limited. To my knowledge, the only relevant study, by Cissé *et al.* (2015), surveyed 300 adults PLHs in Bamako, Kati, and Koulikoro. This study found that HIV stigma (not necessarily related to the legislation) discouraged serostatus disclosure, noting that the law was seen as “adding even more complexity to the issue of disclosure” (p. 3). Therefore, this subsection provides evidence suggesting that the legislation was known to the public during the study period.

I gathered information from local experts, including public health researchers and NGO staff with extensive experience in HIV control programs in Mali. They reported that the law was developed collaboratively with the Ministry of Health, HIV-focused NGOs, and legal consultants, increasing stakeholders’ awareness. Following its enactment, the full text of the law was published in the government’s official newspaper (Journal Officiel de la Republique du Mali, 2006) and disseminated through posters at various HIV/AIDS care organizations.

Local experts recalled that upon the law’s promulgation, information campaigns were conducted on radio and TV to inform PLHs, healthcare providers, and the general public. These media were the primary tools for government communication on HIV-related issues at that time, as indicated by Mali’s national strategic framework (Haut Conseil National de Lutte Contre le VIH/SIDA, 2006). This framework notes that over 60% of Malian women listened to the radio at least once a week, and about 31% had

access to TV news, prompting the government to collaborate with the media in combating HIV/AIDS from 2006 to 2010.⁶

While specific data for Mali in 2006 is lacking, evidence from around that time highlights the importance of radio and TV as information sources. For example, in 2008, radio and TV were the primary sources of HIV information for high school students in northern Mali (Cissoko *et al.*, 2014). Additionally, data from DHSs across 27 sub-Saharan African countries, including Mali's 2006 DHS, show that exposure to radio and TV was positively associated with HIV/AIDS knowledge and preventive behaviors (Westoff *et al.*, 2011). Thus, these media likely played a crucial role in disseminating information about the new law's enactment.⁷

Overall, individuals had sufficient means – primarily radio – to be informed about the legislation at the time of this study. Consistently, three of the five local experts I interviewed separately confirmed that people were aware of the law in June 2006.

3. Conceptual framework

3.1. Model of HIV testing including fear of legal consequences and stigma

HIV stigma, reflecting negative attitudes toward and beliefs about PLHs, is a significant barrier to HIV prevention and treatment (Mahajan *et al.*, 2008; Valdiserri, 2002). The following theoretical model, adapted from Bursztyн *et al.* (2019), elucidates the law's impact while clearly detailing its influence.

Consider a community where PLHs face discrimination, with the population size normalized to one. This community comprises two types of agents with different beliefs about their HIV serostatus based on past sexual behavior: positive (type *H*) or negative (type *L*). The exogenous proportion of type-*H* agents is $q \in (0, 1)$. For simplicity, these beliefs are assumed to accurately reflect their serostatus, although confirmation requires testing. This assumption is plausible; for instance, Arimoto *et al.* (2016) found that factory workers' subjective probabilities of their own serostatus in South Africa predicted their actual HIV status.⁸

Undergoing testing would benefit the agents because their recent serostatus information would lead them to update their sexual behavior (Boozer & Philipson, 2000). Specifically, they would adopt protective measures for themselves and, if they are altruistic, for their sexual partners as well. This expected benefit is assumed to be the same for both types, such that $b_H = b_L = b > 0$, because introducing asymmetry does not add significant insights into this study.

Type-*H* agents are more likely to fear learning their serostatus compared to type-*L* agents. This fear may be exacerbated if the law criminalizes HIV non-disclosure, exposure, and transmission or if it allows healthcare professionals to inform partners of an individual's serostatus. Additionally, type-*H* agents might experience greater anxiety post-testing – one psychological cost of HIV stigma – due to potential harm to their self-

⁶In the 2006 DHS, approximately 64.5% of the female sample listened to the radio at least once a week, and 33.6% watched TV, compared to 77.7% and 47.9% of the male sample, respectively.

⁷In the 2006 DHS, about 87.3% of the female sample could not read a simple sentence at all, and about 93.9% did not read newspapers or magazines weekly. For males, these figures were 67.1% (plus an additional 12.1% who could read only parts of the sentence) and 85.0%, respectively.

⁸Conversely, Anglewicz and Kohler (2009) noted that rural Malawians overestimated their likelihood of infection. However, this also implies that HIV-positive individuals are unlikely to misperceive their actual status.

image. Consequently, type- H agents face a higher total psychological cost for HIV testing than type- L agents ($c_H > c_L \geq 0$), despite identical opportunity costs related to travel and clinic wait times.

For simplicity, it is assumed that those tested are surely and publicly known. Therefore, an agent i (either H or L) incurs an additional cost of HIV stigma if perceived as a PLH by peers (e.g., colleagues, friends, neighbors, and spouses).⁹ Let $\rho > 0$ denote the incremental cost of being perceived as a PLH, and let a_i be a binary indicator for whether agent i has taken the test (1 if yes, 0 if no). Additionally, by writing $c_i = c_H$, I hereafter indicate that agent i is seropositive, as all HIV-positive agents in this model are by assumption type- H . The stigma cost is thus expressed as $\rho \text{Prob}_{-i}(c_i = c_H \mid a_i = 1)$, where $\text{Prob}_{-i}(c_i = c_H \mid a_i = 1)$ represents the peers' (endogenously determined) belief that agent i undergoing HIV testing is seropositive (hereafter, the subscript $-i$ is omitted for simplicity). Agents will choose to undergo testing if the benefits outweigh the costs, represented by:

$$\theta_i > \rho \text{Prob}(c_i = c_H \mid a_i = 1), \quad (1)$$

where $\theta_i \equiv b - c_i$ (implying $\theta_L > \theta_H$).

To illustrate a case of interest, I assume $\theta_i > 0$ and $\theta_L > \rho > \theta_H > 0$; that is, HIV testing benefits agents in the absence of discrimination by peers, and discrimination is primarily a significant concern only for type- H agents. Under these assumptions, all type- L agents would undergo testing, resulting in $\text{Prob}(c_i = c_H \mid a_i = 1) = \frac{q\text{Prob}(a_i=1 \mid c_i=c_H)}{q\text{Prob}(a_i=1 \mid c_i=c_H)+1-q}$ by Bayes' rule. Therefore, if an interior equilibrium, wherein only a fraction of type- H agents undergo testing, is assumed, they must be indifferent about testing vs. not testing; that is, $\theta_H = \frac{\rho q \text{Prob}(a_i=1 \mid c_i=c_H)}{q\text{Prob}(a_i=1 \mid c_i=c_H)+1-q}$. Solving this yields:

$$\text{Prob}(a_i = 1 \mid c_i = c_H) = \frac{1 - q}{\frac{q\rho}{\theta_H} - q} > 0, \quad (2)$$

$$\text{Prob}(a_i = 1) = \frac{1 - q}{1 - \frac{\theta_H}{\rho}} > 0, \quad (3)$$

where $\text{Prob}(a_i = 1) \in (1 - q, 1)$; that is, $q\rho > \theta_H$.

Coercive provisions in HIV-specific legislation increase the incremental cost of being perceived as a PLH (i.e., ρ) by signaling state support for discrimination. This reduces overall testing uptake and particularly among HIV-positive individuals (i.e., $\frac{\partial \text{Prob}(a_i=1)}{\partial \rho} < 0$ and $\frac{\partial \text{Prob}(a_i=1 \mid c_i=c_H)}{\partial \rho} < 0$). This decline occurs when agents are informed about the law through local media (e.g., radio), implying a positive law-induced stigma cost, $\Delta\rho > 0$. The increase in stigma cost is likely greater in rural communities than urban ones ($\Delta\rho_R > \Delta\rho_U$), reflecting stronger social ties and mutual livelihood support (e.g., informal insurance) in rural areas, which intensifies the reduction in rural testing uptake. Similar reductions in testing uptake also occur when the fear of legal consequences (e.g., legal punishment and involuntary serostatus notification) or test-induced self-esteem damage (both constituting c_i) increase due to the legislation, such that $\theta_L > \rho > 0 > \theta_H$,

⁹In psychiatry, this "perceived" or "internalized" stigma, including the abovementioned damage to self-image, contrasts with "enacted" stigma, which refers to actual unfair treatment (Gray, 2002).

resulting in $\theta_H < \rho$ $\text{Prob}(c_i = c_H \mid a_i = 1)$. In summary, *ceteris paribus*, increased HIV stigma due to the legislation is likely to discourage testing uptake.^{10,11}

3.2. HIV testing protocol in the DHS

According to the protocol approved by Mali's National Ethics Committee, the DHS program conducted HIV testing as follows: Field workers first obtained signed consent from respondents, confirming their understanding and willingness. They then collected dried blood spot samples from a finger prick using filter paper, ensuring proper hygiene and safety. Both the blood collection and DHS interview were conducted individually at respondents' homes. The samples were then sent to a laboratory for serostatus testing via enzyme-linked immunosorbent assay (see Macro International, 2007).

In the DHS, HIV testing was conducted solely for surveillance, with no counseling provided during household visits. Field workers, trained for sample collection but not typically medical personnel, collected blood samples. To ensure anonymity, blood samples were identified only by bar codes, and laboratory technicians examined them for HIV antibodies after removing all personal identifiers (except the bar code) from the survey data. Respondents did not see their results, and confidentiality was emphasized during the informed consent process.

In principle, the confidentiality of test results (even to participants) offers no direct benefit to those tested and eliminates the cost of testing uptake. While this might seem to undermine the suitability of this setting for this study's question, this assumption is incorrect for two reasons.

First, the high uptake rates for HIV testing indicate that DHS respondents likely derived positive utility from participating, such as aiding field workers or contributing to public health. This utility is akin to $b_H = b_L = b > 0$, as considered in the model. This is plausible given Mali's active engagement in HIV/AIDS control during the 2006 DHS.¹² Concerns about AIDS were also emphasized during the informed consent process.¹³ This conjecture is also supported by interviews I conducted for this study with two DHS

¹⁰Conversely, protective provisions within HIV-specific legislation – such as confidentiality of test results, counseling, and anti-discrimination measures – reduce the fear of learning one's serostatus (i.e., c_i) and discrimination (i.e., ρ), thereby encouraging testing uptake overall and particularly among HIV-positive individuals if $\theta_L > \theta_H > \rho$ due to the legislation. This is because it suggests $\theta_H > \rho$ $\text{Prob}(c_i = c_H \mid a_i = 1)$. Thus, HIV-specific legislation has two opposing effects on testing uptake.

¹¹This model addresses only the short-term impact of the law on testing uptake. Its long-term effects on HIV prevalence remain uncertain. The law may directly affect sexual behavior, and the combined protective and punitive measures could both encourage and discourage testing. Consequently, test results could influence risky sexual behavior (Baird *et al.*, 2014; Gong, 2015; Thornton, 2008, 2012), leading to ambiguity in the resultant impact on HIV prevalence. Moreover, changes in HIV prevalence would further alter health behaviors (e.g., partner change), affecting HIV prevalence in turn (Kremer, 1996).

¹²In Mali, HIV/AIDS control was formally integrated into both national and local development policies in 2004. The period from 2004 to 2011, marked by a significant increase in international funding, was a peak in the fight against HIV/AIDS (Castro, 2013). In 2004, the government established a comprehensive governance system for HIV/AIDS control, including a national framework, coordination body, and monitoring and evaluation system (Castro, 2013). That year, Mali also declared free access to antiretroviral therapy (Tamí-Maury *et al.*, 2012; Teisseire *et al.*, 2008).

¹³The consent discussion included the statement: "Le sida est une maladie grave, habituellement mortelle. Nous sommes entrain de mener le test de VIH pour mesurer la gravité du problème du sida au Mali" (AIDS is a serious, usually fatal disease. We are conducting the HIV test to assess the severity of the AIDS problem in Mali) (Samaké *et al.*, 2007).

specialists knowledgeable about the program's practices, including the 2006 DHS in Mali. These specialists confirmed that respondents were informed that their survey data could help the government and other stakeholders improve healthcare services, thus providing indirect benefits.¹⁴

Second, concerns about learning one's serostatus or potential legal consequences (such as punishment or involuntary serostatus notification) are not relevant in this setting, as results could not be linked to specific individuals. However, HIV stigma remains a concern if household members could identify tested individuals and speculate that those tested are HIV-positive. According to DHS specialists, while blood collection and interviews were private, the consent process was not always so. Other household members might have observed or overheard the consent process, making it difficult to ensure complete confidentiality given the time required for interviews and blood collection. Consequently, stigma from family members – and possibly from community members who interacted them or were present during the interview – could influence respondents' decisions to accept or refuse HIV testing. This effect would likely have been especially pronounced if respondents misunderstood the testing protocol, particularly regarding the anonymity and confidentiality of test results.

This view aligns with prior studies highlighting the impact of HIV stigma on blood collection during the DHS. Research estimating national HIV prevalence in Africa using DHS data has shown substantial refusal bias (Adegboye *et al.*, 2020; García-Calleja *et al.*, 2006), with those refusing testing more likely to be HIV-positive compared to participants (Marston *et al.*, 2008; Mishra *et al.*, 2008; Reniers & Eaton, 2009). For example, Mishra *et al.* (2008) used multivariate models with common predictors to estimate HIV prevalence among those who refused testing, finding a prevalence of about 3.71% among women who refused compared to 1.83% among those tested, based on 2003 DHS data from Burkina Faso – a neighboring country to Mali (no similar study has been conducted with DHS data from Mali). Although respondents – regardless of their serostatus or knowledge of it – did not gain additional information from participating in HIV testing during the DHS, refusal bias was evident, suggesting that the psychological cost of HIV testing, likely linked to stigma, varies by serostatus (i.e., $c_H > c_L \geq 0$), even when results are strictly anonymous. Alternatively, respondents – especially those who are HIV-positive – might have perceived that confidentiality was not fully assured within households, extended families, or tight-knit communities (i.e., $\rho \text{ Prob}(c_i = c_H \mid a_i = 1) > 0$).

In summary, HIV testing during the DHS differed from typical voluntary counseling and testing at health facilities. The lack of opportunity costs related to visiting and waiting at a clinic, the indirect benefits of participating in blood collection, and the anonymity and confidentiality of results contributed to high uptake rates during the DHS (as detailed shortly).¹⁵ Despite this, the test setting – where HIV stigma remained a concern – still enables this study to evaluate the impact of HIV-specific legislation on voluntary testing through its effect on HIV stigma.

¹⁴For example, during the 2006 DHS, anemia and HIV testing were offered separately. The consent for anemia testing included: “L'anémie qui est due à une alimentation pauvre est un sérieux problème de santé. Les résultats de cette enquête permettront d'aider le gouvernement à mettre en place des programmes pour prévenir et traiter l'anémie” (Anemia due to poor nutrition is a serious health problem. The results of this survey will help the government establish programs to prevent and treat anemia) (Samaké *et al.*, 2007).

¹⁵The uptake rate is notably high considering that only 6% of female and 8% of male respondents reported having been tested voluntarily for AIDS.

4. Data

I use cross-sectional data from the 2006 Standard DHS in Mali, which provides nationally representative information on population, health, and nutrition. The survey employed a two-stage sampling process: communities (clusters) were first selected from the 1998 General Census of Population, followed by household selection within these communities. Women aged 15–49 in all selected households and men aged 15–59 in one-third of the households were interviewed. Additionally, anonymous and voluntary HIV testing was offered to individuals in the households selected for men's interviews (hereafter, referred to as the HIV testing sample).

For the RD design, it is crucial to note that the DHS was conducted from April to December 2006, coinciding with the enactment of Mali's HIV-specific law. I analyze the HIV testing sample who completed their interviews within a 50-day symmetric window around the day of legislation, as determined by data-driven bandwidth selection (see Section 5).¹⁶ The main study sample comprises 1658 women and 1456 men from 148 communities, representing approximately 33% and 34% of the total female and male HIV testing samples, respectively.

The main outcome in this study is an indicator equal to one for respondents who consented to blood collection for HIV testing. The top two panels of Figure 1 display this indicator averaged within 5-day bins (scatter plots) for the main study sample, along with predicted values (solid line) and 95% confidence intervals (dashed line) from a linear polynomial regression, with standard errors clustered at the community and day-of-interview levels.¹⁷

Four findings emerge. First, in the main study sample, 97.7% of women and 95.0% of men agreed to undergo HIV testing. Second, especially for women, there is an upward trend in testing acceptance approaching the day of legislation, suggesting that field workers may have enhanced respondents' understanding and willingness over time. Third, there was a notable decrease of approximately 3–4 percentage points in women's testing uptake on the day of legislation, indicating that the HIV-specific legislation discouraged testing among women only. Fourth, while women's testing uptake appears to have rebounded to pre-legislation levels after this decline, the drop actually persisted for several months post-legislation, as shown in the bottom left panel of the figure, which illustrates longer-term trends using predicted values from a quadratic polynomial regression with 95% confidence intervals.

Regarding the representativeness of the main study sample, Figure 2 illustrates the locations of surveyed communities (marked with crosses) and the mean interview month for each cercle, Mali's second-level administrative unit. Aside from the eastern areas and the northern desert (which was not surveyed), respondents interviewed in June and July were evenly distributed and relatively close to each other. However, the main study sample largely consisted of individuals from poorer rural families, as indicated by measures such as education, health, and household property (see Figure S1

¹⁶ Respondents interviewed before and after this window comprised 9.2% and 57.3% of the female HIV testing sample, and 8.6% and 56.6% of the male HIV testing sample, respectively. On average, 48 respondents eligible for HIV testing were interviewed daily.

¹⁷ The 5-day bins, while arbitrarily chosen, are reasonable given that the data-driven average optimal bin lengths for the left and right sides of the cutoff, which minimize the integrated mean squared error of the binned estimator under evenly spaced partitions (Calonico *et al.*, 2015), are 3.571 and 5.000 for women, and 4.167 and 6.250 for men.

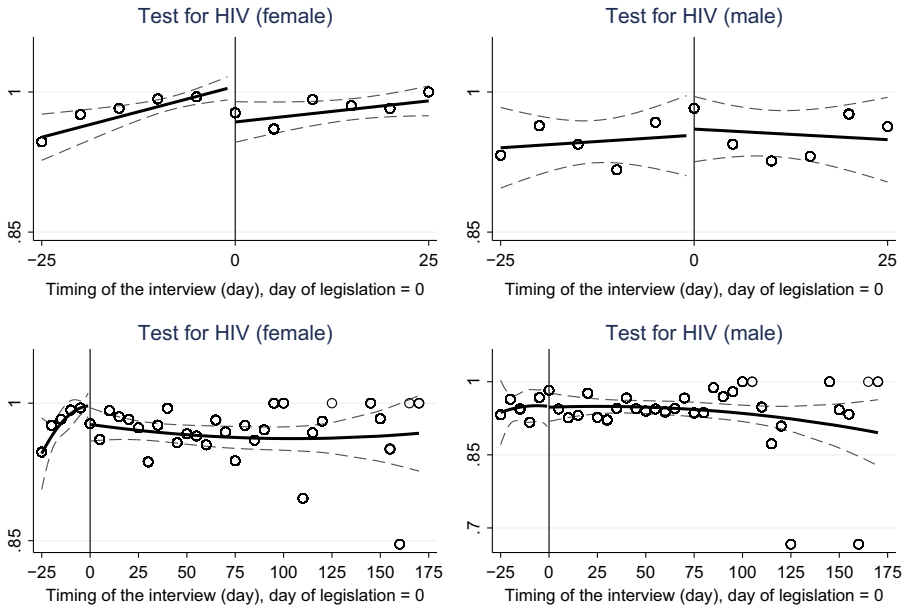


Figure 1. HIV testing uptake.

Note: (1) The top two panels show the value of the analyzed variables averaged within five-day bins (scatter plots), along with the predicted outcomes (solid line) and 95% confidence intervals (dashed line) from a linear polynomial regression of equation (4). The 11 scatter plots represent the mean values for interviews conducted -25 to -21, -20 to -16, -15 to -11, -10 to -6, -5 to -1, 0 to 4, 5 to 9, 10 to 14, 15 to 19, 20 to 24, and 25 days after the legislation. (2) The bottom two panels show the value of the analyzed variables averaged within five-day bins (scatter plots), along with the predicted outcomes (solid line) and 95% confidence intervals (dashed line) from a quadratic polynomial regression of equation (4). (3) In all panels, standard errors are clustered at the community and day-of-interview levels.

of the supplementary material).¹⁸ Therefore, this study's findings should be considered with this sample limitation in mind.¹⁹

5. Empirical strategy

To identify the impact of the HIV-specific legislation, I use the following specification for respondent i :

$$y_i = \alpha_1 + \alpha_2 S_i + \alpha_3 f(d_i - z) + \varepsilon_i, \quad \forall d_i \in [z - h, z + h], \quad (4)$$

where y_i represents the outcome of interest; d_i is the day of the respondent's DHS interview; z is the cutoff day (June 29, 2006); h denotes the bandwidth in days; $S_i = 1(d_i \geq z)$ is a treatment indicator; and ε_i is the error term. The function $f(d_i - z)$

¹⁸This figure compares those interviewed within the 50-day symmetric window to those outside it.

¹⁹If individuals with higher education levels show less discrimination against PLHs (Corrigan and Watson, 2007), the law's impact on the broader population may be less pronounced than what is observed in this study.

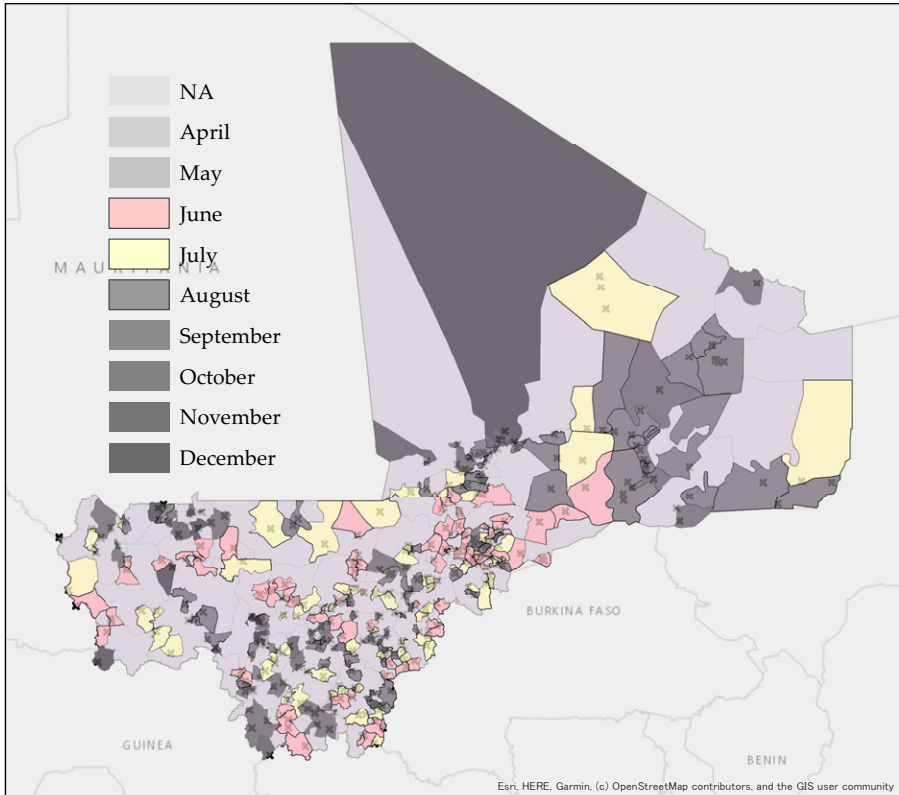


Figure 2. Distribution of the timing of the interview.

Note: This figure shows the locations of surveyed communities (marked with crosses) and the mean interview month for each circle, Mali's second-level administrative unit. I matched the DHS community's GPS coordinates with the country map from the United Nations Office for the Coordination of Humanitarian Affairs (<https://data.humdata.org/dataset/cod-ab-mli?>).

represents an RD polynomial of the time distance $d_i - z$, with separate polynomial coefficients for each side of the cutoff. I estimate equation (4) using a first-order polynomial with $h = 25$ by ordinary least squares (OLS), as the optimal bandwidths minimizing mean squared error of the point estimator for HIV testing uptake were 29.2 days for females and 21.5 days for males (Calonico *et al.*, 2014).

The influence of HIV stigma on testing uptake is likely to vary across communities. Additionally, given the specification error inherent in RD models with a discrete running variable, the model error may be correlated within the interview day (Lee & Card, 2008). Therefore, I cluster standard errors at the community and day-of-interview levels (148 and 51 groups, respectively).

There are two main concerns regarding the validity of the RD design. First, the DHS data exclude individuals who either actively or passively (e.g., absence) refused to be interviewed, which could bias estimates if HIV-specific legislation influenced their

selection into the interview and HIV testing.²⁰ Although the 2006 DHS reported high response rates of 91% for men and 97% for women (Samaké *et al.*, 2007, pp. 9–10),²¹ the RD design still faces this concern. Second, survey enumerators might have rescheduled interviews based on respondents' health characteristics, potentially due to anticipated effects of HIV legislation or by chance (e.g., failing to reach respondents who were away from home during the initial visit and scheduling callback visits).²² For example, interviews of HIV-positive individuals might have been postponed from just before to just after the legislation, while those of HIV-negative individuals could have been moved in the opposite direction.

To mitigate these concerns, I conducted two exercises. First, particularly to tackle the issue of selective sorting, the top left (for women) and top right (for men) panels of Figure 3 show the number of respondents interviewed each day, revealing no clear sorting at the cutoff. Additionally, I applied the density test proposed by Cattaneo *et al.* (2020), which is less restrictive than the seminal test by McCrary (2008).²³ The null hypothesis of no discontinuity was not rejected, as indicated by the p-values reported at the bottom of each panel.

Second, I assessed whether several variables (see Table 1 for summary statistics), likely unaffected by the law, are balanced at the treatment threshold. Specifically, I estimated a local linear polynomial model of equation (4) with $h = 25$ for each variable and reported the standardized coefficient of α_2 with 95% confidence intervals in the bottom panels of Figure 3. The variables examined include individual-level covariates such as age, religion (Muslim), ethnicity (Bambara, Mali's largest ethnic group), highest level of education, height and weight (available only for women), HIV-related knowledge score (0–10, detailed in Section S.1 of the supplementary material), marital status, and circumcision-related indicators (available only for women) as well as an indicator for recent contraction of sexually transmitted diseases (STDs).²⁴ Household-level covariates include household size and ownership of a radio, TV, bicycle, or electricity (indicators). Community-level covariates include an urban indicator, population density, precipitation, temperature, altitude, slope, and distances from Bamako (the capital) and the nearest national border (see Section S.1 for details).²⁵ The variables varied smoothly at the cutoff.²⁶ The subsequent empirical results remain robust when controlling for these covariates.

²⁰Angotti *et al.* (2009) notes that some respondents in Malawi avoided HIV testing by leaving their homes, a phenomenon termed "HIV testing refusal."

²¹I cannot verify these response rates in the public data, as it excludes respondents who refused to be interviewed.

²²During the DHS, field workers typically make up to three callback visits, either at different times on the initial day or on subsequent days; refer to the survey manuals for details.

²³While this non-parametric density test is theoretically less robust due to the discreteness of the running variable (Lee and Card, 2008), it is practically useful if the discrete variable's distribution is not excessively coarse (Lee and Lemieux, 2010, p. 336).

²⁴These indicators are binary: one for respondents who had heard of circumcision, one for those who were circumcised (with zero assigned to uncircumcised individuals and those who had never heard of circumcision), and one for those who had contracted any STDs in the last 12 months.

²⁵Distances are contextually relevant: expected law enforcement may decrease with distance from the capital, and labor migration to neighboring countries (especially Côte d'Ivoire) is common, where migrant workers often engage in risky sexual behavior (Oster, 2012).

²⁶Although statistically insignificant, the bottom panels of Figure 3 indicate a decline in "STDs in 12 months," a known risk factor for HIV infection. This might suggest under-reporting of STDs due to fear of legal consequences, based on a misunderstanding of the legislation, potentially complicating the use of STD

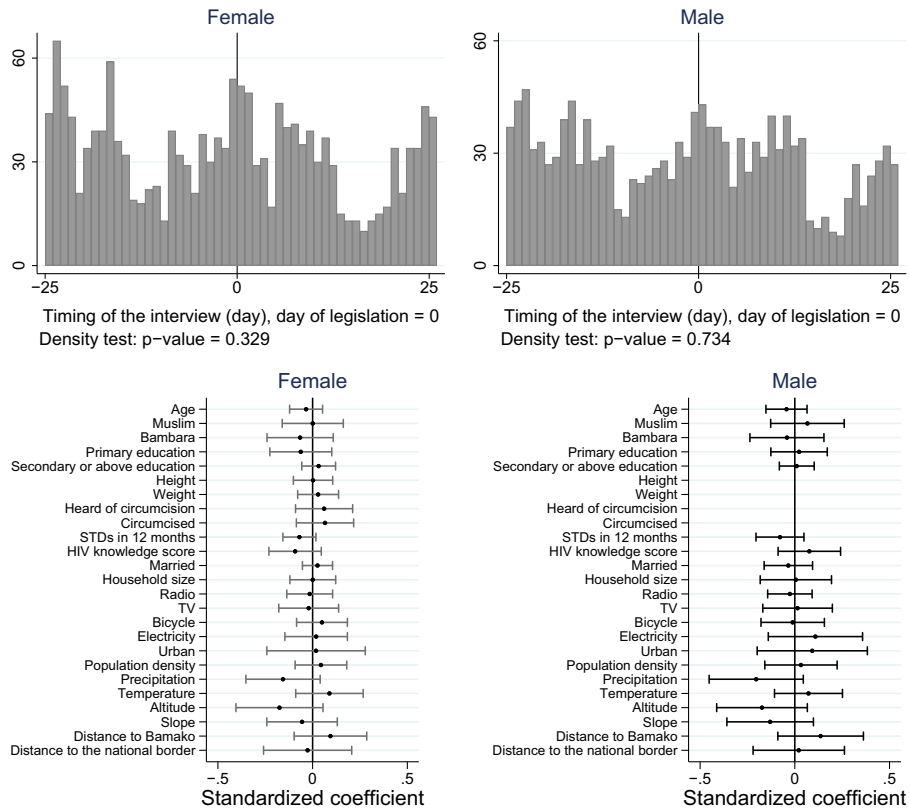


Figure 3. RD validity checks: The number of respondents and balanced covariate checks.

Note: (1) The top two panels display the number of respondents eligible for HIV testing and interviewed within a 50-day symmetric window around the day of legislation. (2) The bottom two panels show the standardized treatment effect on each analyzed variable with 95% confidence intervals, derived from a linear polynomial regression of equation (4). Standard errors are clustered at the community and day-of-interview levels. (3) Height, weight, and circumcision-related information are not available for men.

Finally, the National Assembly passed the bill on June 2, and it went into effect on June 29, coinciding with President Touré's signing of the bill into law. Therefore, one may concern that the bill's passage had an impact before the enforcement. However, according to Mali's Constitution (Article 40), the President must promulgate laws within 15 days of receiving the final text. During this period, the President can request the National Assembly to reconsider certain provisions, which can delay promulgation. The President's signature was thus essential for enforcing the HIV-specific law and signaled the state's official endorsement. Moreover, note that the main analyses use a 50-day symmetric window around the legislation date (June 29), thereby automatically

contraction as a control variable. However, this control has a minimal impact on the empirical findings. Alternatively, the proportion of individuals with fewer STDs—hence a lower risk of HIV—might have increased immediately following the legislation. This potential increase attenuates the law's test-discouraging effect, as those with fewer STDs would face lower costs for HIV testing and thus be less likely to refuse it (see subsection 3.1).

Additionally, concerns about the accuracy of self-reported sexual behavior (Corno & de Paula, 2019) further complicate the analysis.

7. Conclusion

HIV criminalization is a global phenomenon: 82 countries – covering 111 jurisdictions, including subnational entities within Mexico, Nigeria, and the US – have enacted HIV-specific criminal laws (Symington & Bernard, 2022). Sub-Saharan Africa has the largest number of such jurisdictions (30 countries), followed by Eastern Europe and Central Asia, Latin America and the Caribbean, the Asia-Pacific region, the Middle East and North Africa, and North America (only the US) (*ibid.*).³⁷ Public health experts and advocacy groups have expressed concern that these laws reinforce discrimination against PLHs, undermine prevention efforts, and exacerbate the epidemic. The Global AIDS Strategy 2021–2026 (UNAIDS, 2021) sets a target that, by 2025, fewer than 10% of countries will maintain punitive legal and policy environments that result in the denial or restriction of access to healthcare services. Nevertheless, only a few countries have thus far repealed these laws. Despite ongoing global policy discussions, rigorous empirical studies evaluating the impacts of HIV criminalization remain limited.

In this study, I examined the effect of Mali's HIV-specific law on voluntary testing, focusing on HIV stigma. I used an RD design within a unique context: Mali enacted the law during a nationally representative survey offering anonymous and voluntary testing. This setting was ideal for studying stigma rather than legal consequences, as testing was confidential, though household members could still infer who was tested and potentially discriminate based on suspected positive serostatus.

Empirical analyses showed that the law reduced women's HIV testing uptake by approximately 3–4% compared to the day before. Rural women more strongly refused testing than urban women, likely due to higher stigma costs in rural areas. These effects were more pronounced among those with radios, suggesting that radio broadcasts heightened public awareness of the legislation. Women without completed formal education were particularly deterred from testing, likely because they misunderstood the anonymity and confidentiality of test results, thereby experiencing greater stigma-related concerns. In contrast, the law did not impact men's testing uptake. Malian women, who typically depend on men for financial support (Heath *et al.*, 2020; Lees *et al.*, 2021; Van de Walle, 2013), are vulnerable to HIV-related mistreatment from family members. This likely makes women more reluctant to test due to fear of assumed positive serostatus and subsequent mistreatment. The law may have intensified this fear, resulting in a stronger deterrent effect on testing among women than men. This aligns with the common view that HIV criminalization disproportionately affects women, who often first discover their seropositivity through antenatal or other healthcare services and face discrimination from peers who may blame them for the virus. This fear of discrimination likely deterred women from seeking healthcare post-legislation.

The theoretical model also suggested that the law may have deterred testing among HIV-positive individuals. Consequently, HIV criminalization might drive at-risk populations into hiding, potentially increasing HIV prevalence over time. However, due to the small number of seropositive cases and limited data variation, the law's effect on testing uptake among HIV-positive individuals remains inconclusive. Future research should explore this further.

³⁷No country in Western and Central Europe has HIV-specific criminal laws.

RESEARCH PAPER

Rural electrification and fertility decline in Iran

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Abstract

A growing body of evidence finds that rural electrification reduces fertility, typically by expanding women's opportunities outside the home and raising the opportunity cost of childbearing. We examine electrification in post-revolutionary rural Iran, where electricity expanded rapidly but female labor force participation remained low. Using a large panel of villages observed in the 1986, 1996, and 2006 censuses, we show that while Ordinary Least Squares estimates align with the broader literature in suggesting a negative association between electrification and fertility, instrumental variable estimates exploiting elevation-based variation reveal the opposite: villages with longer exposure to electricity experienced higher fertility. This positive effect is strongest in provinces with lower female labor force participation, indicating that the substitution channel emphasized in prior research was weak in the Iranian context. These findings highlight the importance of context in shaping demographic responses to infrastructure and suggest that electrification's effects on fertility are not universally negative.

Keywords: electrification; fertility; impact evaluation; Iran

JEL classification: J13; H54; O18

1. Introduction

Two major developments following Iran's 1979 Islamic Revolution shaped the country's economic trajectory: the rapid expansion of basic infrastructure – roads, electricity, and clean water – to rural areas, and a sharp decline in fertility. These transformations profoundly impacted the living standards of ordinary Iranians. Investment in infrastructure contributed to a reduction in overall poverty levels (Salehi-Isfahani 2009a), while the decline in fertility enabled families to shift their focus from childbearing to human capital investment. As a result, educational attainment – particularly for rural women – increased significantly, and the gender gap in education closed (Assaad *et al.* 2019).

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It is tempting to interpret the simultaneous expansion of infrastructure and the subsequent fertility decline in rural Iran as causal evidence that infrastructure investment reduces fertility, a relationship generally supported by the empirical literature. However, in this paper, we challenge this interpretation using Iranian data. We present evidence from a large sample of 45,000 villages in Iran, which allows us to instrument for electricity availability. While our Ordinary Least Squares (OLS) estimates align with the general negative relationship found in the literature, accounting for the endogenous nature of electricity access using an instrumental variable (IV) – specifically, the deviation of a village's elevation from its district mean – to isolate variation in electricity access unrelated to fertility itself reveals a positive association between electrification and fertility. Our instrument is similar to those used in the electrification literature (e.g., Dinkelman (2011) and Grogan (2016)). It satisfies the relevance condition because it is correlated with the cost of extending electricity to the village and the exogeneity condition because it does not directly affect fertility.

At first glance, the finding of a seemingly counterintuitive positive effect may appear inconsistent with economic theory. However, theory does not predict a universal direction of impact (Vidart 2024). Electrification can influence fertility through competing mechanisms. First, rising incomes and the availability of labor-saving household appliances may increase demand for children (Greenwood *et al.* 2005). Second, expanded market opportunities for women generally raise the opportunity cost of childbearing, causing fertility to decline. Third, improvements in child survival and reduced education costs due to electrification could lead parents to invest more in child quality rather than quantity. Whereas the latter two lower fertility, as mainstream theory suggests, the first raises it.

Post-revolutionary Iran offers a particularly useful context for exploring the fertility effects of electrification, due to historically low – and persistently limited – female labor force participation (LFP) rates in both rural and urban areas. Rural women, in particular, rarely engage in market work outside the home, implying that a key mechanism through which electrification typically reduces fertility – by increasing the opportunity cost of women's time – is largely absent. Following the revolution, while rural economic development was prioritized, strict gender segregation policies in public spaces and workplaces further discouraged female employment (Karshenas and Moghadam 2024; Shahrokni 2019). According to the World Bank, women's LFP in Iran is only 14%, far below that of men (around 90%). Our estimates using a 2% sample of the 1996 census reveal even lower rates: 10.0% in urban areas and 11.5% in rural areas, with some provinces exhibiting female participation rates as low as 5%. We explore the implications of low female LFP for the impact of electrification in section 7.

Since theory is ambiguous on the relationship between electrification and fertility, it becomes an empirical question, one that existing literature explores extensively but inconclusively. While a large body of literature examines this relationship, most studies rely on simple correlations. Using OLS without addressing the endogeneity of electricity access, several studies report a negative relationship: Harbison and Robinson (1985) in six countries, Cornwell and Robinson (1988) in southern U.S. states, and Peters and Vance (2011) in rural Côte d'Ivoire. However, some studies find the opposite, sometimes even within the same country. Cornwell and Robinson (1988) document a positive association in northern U.S. states, as do Peters and Vance (2011) in urban Côte d'Ivoire. In an influential study, Greenwood *et al.* (2005) use structural estimation to argue that electrification contributed to the post-World War II U.S. baby boom. However, Bailey and Collins (2011) challenge this conclusion, arguing that the diffusion of household

appliances and the associated productivity gains occurred decades before the baby boom. A small number of studies addressing electricity's endogeneity all report negative effects – Grogan (2016) in Colombia, Fujii and Shonchoy (2020) in Bangladesh, and Vidart (2024) in the U.S. – primarily contexts where expanded electricity also expanded female labor market opportunities. A cross-country panel study by Belmin *et al.* (2022) reaches a similar conclusion. To our knowledge, our paper is the first to document a positive causal relationship using instrumental variables, highlighting the critical importance of socioeconomic and institutional contexts.

Importantly, finding a positive relationship does not imply that electrification did not contribute to overall fertility decline – even in rural Iran. A key driver of rural fertility decline – the expansion of rural health clinics (Abbasi-Shavazi *et al.* 2009; Salehi-Isfahani *et al.* 2010) – was itself dependent on electricity. Because of this dependence, our IV estimates capture a net effect: the fertility-increasing channel of electrification, partly offset by the fertility-reducing channel through clinics, as noted in Section 7.2. As with all major social changes, multiple factors contributed to fertility decline in Iran. In the case of Iran, beyond the provision of infrastructure and family planning services, the revolution altered incentives for childrearing in other ways. In particular, the government's pro-poor outlook, emphasizing the poor's welfare raised their expectations of returns to investment in child education, encouraging lower fertility.

To minimize bias in our estimates, we control for other potential influences on fertility to the extent that our data permit. Electrification can increase access to education, so we control for the years of exposure to primary and middle schools. Notably, most villages in our sample already had primary schools *before* receiving electricity, implying that electricity's effect through primary school access is not a major concern. We also control other key factors that can influence fertility such as village size, presence of TV and mosque, and whether the village is in the plains or a mountainous area. We also examine the impact of other correlated infrastructure, such as clean water and asphalt roads, and confirm that the positive impact of electrification remains robust even after controlling for these factors.

Our exclusive focus on electrification in this paper is motivated by two considerations. First, the timing of electricity extension to villages is more easily pinpointed than that of roads or clean water infrastructure. Second, among all types of infrastructure, electrification is most closely related to women's opportunity cost of time, and therefore fertility.

Since electricity in Iran is publicly produced and provided, its placement in villages is likely not random. The government may have prioritized locations with lower placement costs, favoring areas that are less mountainous and easier to access. Ignoring elevation as a determinant of electricity placement would bias the estimated effect of years of exposure to electricity on fertility. This omitted-variable bias would likely be downward, as electrification was prioritized in more accessible, lower-fertility villages, potentially masking a true positive effect. Indeed, consistent with other studies cited above, our OLS estimates show a negative relationship. It is only when we instrument for placement of electricity that we find a positive impact.

The plan of the paper is as follows. The next section describes the Iranian context and how the electricity and fertility transitions took place. Section 3 describes the village sample and the instrument we employ. Section 4 discusses our methodology, and Section 5 presents and discusses the results. Section 6 discusses the importance of female LFP as the mechanism through which electrification affects fertility, and Section 7 checks for the robustness of the results in view of potential sources of bias, such as

variation in child mortality, migration, and other infrastructure that affect child-woman ratios (CWRs). Section 8 concludes.

2. Iranian context

In this section, we describe the two major developments that underpin our study – rural electrification and fertility decline – both of which unfolded after the Islamic Revolution. As a popular revolution, it was natural that it would be followed by the rapid extension of infrastructure and basic services to rural and other disadvantaged areas (Lob 2020; Salehi-Isfahani 2009b). Within a decade, most villages were connected to the national electricity grid and gained access to clean water.

Yet, as a conservative revolution, fertility decline was not part of the original plan. In fact, as we discuss below, the new regime's early rhetoric and policies were explicitly pro-natalist, and only shifted once the urgency of family planning became clear. This policy reversal, combined with the revolution's pro-poor orientation and emphasis on education, contributed to a rapid decline in fertility.

2.1. Family planning and rural fertility

Arguably, fertility decline also had its roots in the revolution, but in an indirect way. In its early years, the revolutionary state sought to eliminate Western influences by dismantling the Shah's family planning program, halting the distribution of contraceptives, and promoting large families. Within a decade, however, the contradiction between pro-natalist policies and the government's simultaneous commitment to eliminating illiteracy and providing universal education resolved in favor of the latter. A new, rural-focused family planning program was launched in 1989, later earning global recognition for its innovation and setting what has been described as a world record for fertility decline (The Economist 2009).

Although this rapid decline is often attributed to the family planning program (Abbasi-Shavazi *et al.* 2009), its direct contribution has been estimated at less than 20% of the decline (Salehi-Isfahani *et al.* 2010). Clearly, other changes – including the expansion of infrastructure – also played an important role, but these influences have received little systematic attention. In particular, the effect of electrification, which preceded the family planning program, has not been rigorously examined. Our analysis does not diminish the importance of family planning; rather, it shows that the direct effect of electrification was to raise fertility. At the same time, as we discuss in Section 7.2, electrification also contributed indirectly to fertility decline by enabling the expansion of family planning centers.

At the time of the revolution, rural fertility was at its peak – about eight births per woman. Urban fertility was also high, though consistently lower than rural fertility (Figure 1). Beyond these high levels, the most important feature of fertility in the early years of the revolution was the government's explicitly pro-natalist orientation. The revolutionary state dismantled the Shah's family planning program immediately after coming to power, dismissing it as a colonial legacy. This stance was further reinforced during the war with Iraq (1980–1988), when intense ideological mobilization and the loss of many young men strengthened the regime's encouragement of large families.

After the war ended, the government's priorities shifted from sustaining the war effort to fulfilling its promise of improving living standards, particularly for the poor, whom it regarded as its social base. Results from the 1986 census revealed rapid

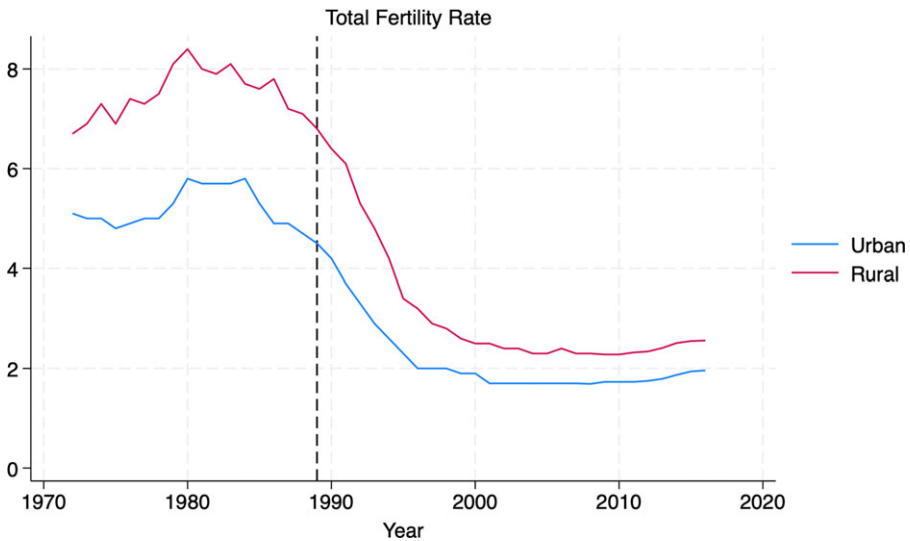


Figure 1. Fertility decline in rural and urban areas.

Source: Abbasi-Shavazi M. J. (2020)

population growth of 3.9%, and overcrowding in elementary schools threatened the regime's objective of universal education. Enrollment in first-grade classes had risen by 40% – far outpacing the construction of new schools – forcing some schools to operate in three shifts per day (Salehi-Isfahani 2005).

In a surprising but pragmatic move, the conservative fundamentalist regime reversed its pro-natalist stance and adopted family planning. The rural health network was commissioned with a new mandate: to reduce the birth rate (Abbasi-Shavazi *et al.* 2009; Salehi-Isfahani *et al.* 2010). The program achieved extraordinary success – acceptance of modern contraceptives among rural families eventually exceeded that of urban families – largely because the religious government enjoyed the trust of rural households. This trust was built on the clinics' prior role in providing mother and child health care before promoting birth control. In this way, the program succeeded where the Shah's had failed: it managed to reduce rural fertility (Aghajanian 1995). Importantly, however, the expansion and functioning of the rural health network depended on electrification, making electricity not only a direct influence on fertility but also an indirect contributor to fertility decline through family planning.

As part of this program, the government altered incentives for limiting family size by eliminating subsidies for children beyond the third. Yet whether fertility decline was driven by family planning, these new incentives, or other factors remains unclear. Salehi-Isfahani *et al.* (2010) estimate that rural clinics accounted for less than 20% of the overall decline. Economists are generally skeptical that governments can exert large influence on deeply ingrained behaviors such as childbearing. Instead, they emphasize the role of economic development and education in shaping fertility preferences (Schultz 1981; Pritchett 1994; Miller 2010; Miller and Babiartz 2016). Iran's own experience, as illustrated in Figure 1, supports this view: fertility decline in rural areas had already begun before the implementation of family planning.

Table 1. Distribution of villages in the full sample by year of electrification

Year of obs	Electrified villages		Population with electricity	
	Number	Percent	Million	Percent
1966	666	1.5	0.8	4.1
1973	1392	3.1	1.8	9.4
1976	2659	5.9	3.4	18.0
1981	10049	22.1	8.9	47.2
1986	17456	38.4	12.5	66.2
1988	20133	44.3	13.6	72.1
1996	31461	69.3	17.0	90.0
2006	41915	92.3	18.8	99.2
Observations	45,419	100	18.9	100

Note: Villages in the full sample, observed in the censuses of 1986, 1996, and 2006. Village populations are in 2006.
Source: National Censuses of Population, Statistical Center of Iran.

2.2. Electrification

Replacing a western-oriented, urban-biased monarchy, for its first two decades the Islamic Revolution shifted investment to rural areas. The Reconstruction Crusade, which is now part of the Ministry of Agriculture, mobilized thousands of young volunteers nationwide to build roads and extend electricity and clean water to villages (Lob 2018). Census data show that between 1966 and 1976, during a decade of record economic growth, the percentage of electrified villages increased from 1.5% to 5.9%, and the share of the population living in them from 4.1% to 18.0% (Table 1). These are modest increases in comparison with urban areas that had achieved full electrification by 1980 (Figure 2). Drawing on annual household surveys, Figure 2 shows the rapid expansion of electrification in rural areas, which closed the gap with urban areas by 2010. By 1981, the percentage of electrified villages and the population that lived in them (as of 2006) had increased to 22.1% and 47.2%, respectively. During the oil boom years (1973–1976), each year 422 villages were electrified, whereas during the Iraq War (1981–1986), this number jumped to 1,481 villages. By 2006, 92.3% of villages and 99.2% of the population living in them had access to electricity.

3. Data

The unit of observation in this study is the village. Our data come from three consecutive population censuses – 1986, 1996, and 2006 – which provide basic information on all Iranian villages. We focus on these years because they span the period of rapid electrification and fertility decline at the center of our analysis.

Each census enumerates roughly 100,000 villages, many of which are very small hamlets or temporary settlements. To construct a consistent panel, we restrict attention to the 43,609 villages that appear in all three censuses with a non-zero population. We refer to this set as the full sample, and its summary statistics are reported in Table 2.

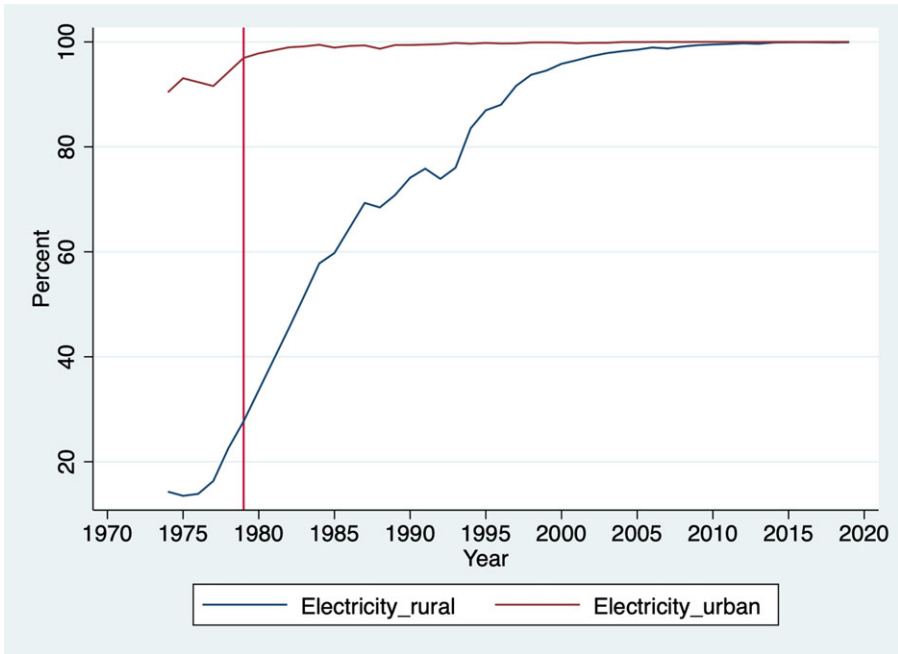


Figure 2. Rapid expansion of rural electrification after the revolution (percent of population with electricity).

Source: Household Expenditure and Income Surveys, Statistical Center of Iran, various years.

These censuses provide basic demographic information and community characteristics, including topography, availability of schools, access to basic services, and the presence of a mosque.¹ For our fertility measure, the CWR, the Statistical Center of Iran supplied village-level counts of children under age five and women aged 15–49 (see Salehi-Isfahani *et al.* (2010) for details).

In 1986, the average CWR was 1.0, roughly corresponding to a total fertility rate (TFR) of about seven births per woman. By 1996, the average CWR had dropped to 0.56, and by 2006 it had further declined to 0.31, implying a TFR just above replacement level. Some villages report CWR values outside the plausible range (e.g., the maximum CWR in Table 2 is 7.5, which implies a biologically impossible TFR). These outliers likely reflect unusual population structures; excluding them in sensitivity checks below does not affect our regression results.

A limitation of the CWR is its sensitivity to infant and child mortality. Electrification can raise the CWR if it reduces child mortality and allows more children to survive to age five, which enters the numerator of the ratio. Because the village-level census data do not include information on infant or child mortality, we rely on province-level estimates of under-five child mortality rates (CMR) for 1990 and 2019. From these benchmarks, we construct province-level CMR measures for the census years used in our study. In Section 7.1, we incorporate these measures, together with other proxies for child

¹A related dummy variable identifies whether a village is Shia- or Sunni-majority. About 90% of Iranians belong to the Shia sect of Islam.

Table 2. Summary statistics, full sample

	Mean	Std. Dev.	Min	Max
Child-woman ratio 1986	1.004	0.361	0.031	7.5
Child-woman ratio 1996	0.560	0.281	0.025	5
Child-woman ratio 2006	0.312	0.182	0	3
Electricity exposure (to 1996)	7.930	7.730	0	30
Electricity exposure (to 2006)	15.961	10.065	0	40
Elevation	1.330	0.681	-0.100	2.951
Village population 1986	395	575	2	22792
TV exposure in 1996	0.749	0.433	0	1
Primary school exposure (to 1996)	20.495	12.145	0	46
Middle school exposure (to 1996)	2.346	6.396	0	30
<i>Proportion of villages with</i>				
Mosque	0.660	0.474	0	1
Shia majority	0.748	0.434	0	1
<i>Village geography</i>				
Plain	0.420	0.493	0	1
Mountain	0.563	0.496	0	1
Forest	0.017	0.131	0	1
No. of observations	43,609			

Notes: The number of observations for the child-woman ratio in 2006 is 39,474 and for elevation is 20,274.

mortality, as controls in the regressions to reduce potential bias from the missing mortality variable.

We measure years of exposure to electricity from the earliest census year in which a village is recorded as having access. This does not necessarily correspond to the exact year electrification occurred, since the data report availability only in the census years and other national reports listed in Table 1. As a result, the distribution of exposure years is somewhat irregular. Exposure is measured up to 1996 and 2006, with average village exposure equal to 7.9 years in 1996 and 16.0 years in 2006.

As noted earlier, to address potential endogeneity of electricity exposure, we turn to instrumental variable (IV) estimation. Our instrument is a geographic proxy for the cost of electrification, defined as the difference between a village's elevation above sea level and the mean elevation of its district. The idea is that elevation is correlated with the cost of extending rural electricity but not directly with fertility. Using relative elevation, rather than absolute elevation, further ensures that we capture variation in electrification costs within districts while netting out broader differences in geography, climate, and agricultural potential that might otherwise be correlated with fertility. Iran's mountainous geography – with villages located in three main zones: mountain, forest, and plain (Fisher and Boyle 1968; Modarres and da Silva 2007) – provides a suitable basis for using elevation as a proxy for electrification costs.

Table 3. Summary statistics, IV sample

	Mean	Std. Dev.	Min	Max
Child-woman ratio 1986	0.980	0.312	0.031	7.5
Child-woman ratio 1996	0.516	0.227	0.025	4
Child-woman ratio 2006	0.279	0.140	0	2
Electricity exposure (to 1996)	9.172	7.632	0	30
Electricity exposure (to 2006)	17.966	9.303	0	40
Elevation	1.330	0.681	−0.100	2.951
Village population 1986	469	561	5	7545
TV exposure in 1996	0.818	0.385	0	1
Primary school exposure (to 1996)	23.717	10.419	0	46
Middle school exposure (to 1996)	2.965	6.972	0	30
<i>Proportion of villages with</i>				
Mosque	0.734	0.442	0	1
Shia majority	0.863	0.343	0	1
<i>Village geography</i>				
Plain	0.463	0.499	0	1
Mountain	0.522	0.500	0	1
Forest	0.015	0.121	0	1
No. of observations	20,274			

Since village elevations are not reported in the census, we obtained elevation data from the website of the National Cartographic Center of Iran. Merging these data with our village sample reduces the IV sample to 20,274 observations – less than half of the full sample. Fortunately, the distributions of the key variables are very similar across the IV and full samples, suggesting that the loss of observations does not materially affect the representativeness of our analysis.

Table 3 reports summary statistics for the IV sample. The average CWR was 980 children per 1,000 women in 1986, falling to 516 in 1996 and 279 in 2006. The table also provides information on years of exposure to electricity, population size, years of exposure to primary and middle schools up to 1996, and religious characteristics in 1986. The average village population in 1986 was 469 persons, or roughly 90 households. By 1996, villages had been electrified for an average of 9 years, whereas exposure to primary schools was much longer, averaging 23.7 years. The data indicate that most villages received access to middle schools after electrification. Finally, TV coverage data, available only for 1996, show that about 80 percent of villages had television access by that year.

Religious characteristics in 1986 show that 73% of villages had a mosque – an indicator of both local wealth and religiosity – and 86% had a Shia majority. Geographic data indicate that 46% of villages were located in plains, 52% in mountainous areas, and only 2% in forests, underscoring Iran's predominantly mountainous terrain.

Table 4. Fertility and electricity exposure by province

	CWR 1986	CWR 1996	CWR 2006	Exposure 2006
Markazi	0.828	0.398	0.222	21.5
Gilan	0.737	0.390	0.200	18.8
Mazandaran	0.749	0.346	0.213	23.4
East Azerbaijan	0.975	0.519	0.311	14.5
West Azerbaijan	0.925	0.588	0.330	14.6
Kermanshah	0.992	0.541	0.279	17.0
Khuzestan	1.149	0.735	0.404	14.8
Fars	1.111	0.509	0.285	15.5
Kerman	1.060	0.572	0.338	18.4
Khorasan Razavi	0.975	0.542	0.321	17.4
Isfahan	0.939	0.425	0.251	25.5
Sistan	1.220	0.716	0.556	8.6
Kurdestan	0.996	0.597	0.288	14.9
Hamadan	1.013	0.528	0.261	20.0
Bakhtiari	1.201	0.708	0.384	15.4
Lorestan	1.113	0.638	0.330	14.2
Ilam	1.140	0.568	0.263	19.4
Kohkiluyeh	1.251	0.686	0.357	11.5
Bushehr	1.001	0.475	0.298	18.9
Zanjan	0.982	0.607	0.288	14.8
Semnan	0.765	0.421	0.277	22.5
Yazd	1.011	0.472	0.295	22.8
Hormozgan	1.088	0.723	0.433	12.5
Tehran	0.709	0.427	0.275	25.4
Ardebil	1.036	0.541	0.280	12.8
Qom	0.858	0.457	0.281	24.4
Qazvin	0.914	0.462	0.238	17.7
Golestan	0.998	0.537	0.296	26.7
North Khorasan	0.978	0.608	0.317	13.0
South Khorasan	0.900	0.577	0.334	9.7
Total	0.998	0.553	0.312	16.6

Note: Villages in the full sample, observed in the censuses of 1986, 1996, and 2006.

Table 4 reports average years of exposure to electricity and the CWR for 30 Iranian provinces. The contrast between more advanced provinces (e.g., Isfahan, Gilan, and Tehran) and disadvantaged ones (e.g., Sistan, Kohkiluyeh, and South Khorasan) is

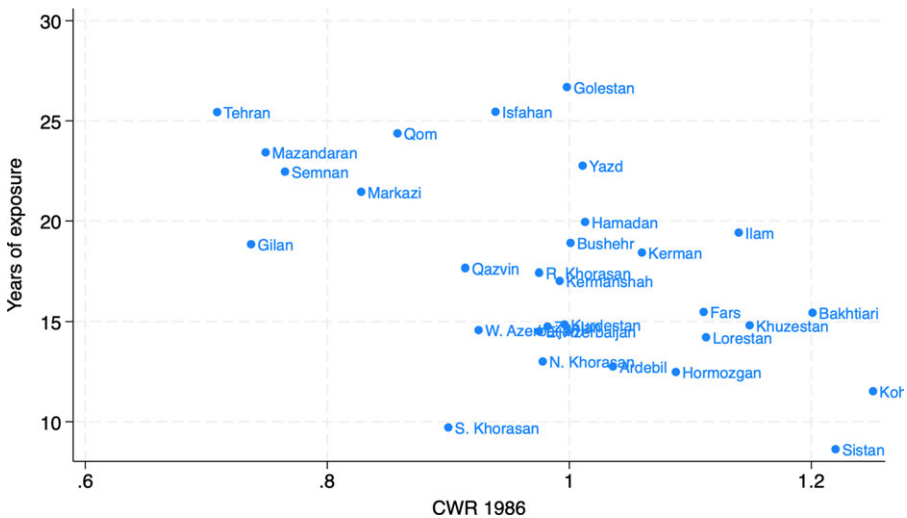


Figure 3. CWR and years of exposure to electricity by province.

evident in both electrification exposure and fertility decline. Province-level averages reveal substantial variation in both outcomes. By 2006, several advanced provinces – notably Tehran and Isfahan – had accumulated more than 20 years of electrification, while disadvantaged provinces averaged as little as 8.6 years. In 1986, provincial CWRs ranged from 0.709 in Tehran to 1.251 in Kohgiluyeh, and by 2006, from 0.200 in Gilan to 0.556 in Sistan.

Province-level variation in fertility is positively correlated with years of electrification exposure: the correlation between CWR in 1986 and exposure by 2006 is 0.57. This relationship is also illustrated in Figure 3, which places Iran's least developed provinces (Sistan, Kohgiluyeh, and Bakhtiari) at one extreme – with high fertility in 1986 and limited exposure – and the most developed provinces (Tehran, Gilan, and Isfahan) at the other. Interpreting CWR in 1986 as a proxy for provincial development, the pattern suggests that less developed provinces were electrified later.

4. Methodology

Our baseline OLS specification estimates the following equation, expressing CWR as a function of years of exposure to electricity and other village characteristics:

$$Y_i = \alpha + \beta E_i + X_i \psi + \varepsilon_i, \quad (1)$$

where Y_i is the CWR in 1996 or 2006 for village i , E_i is the number of years of exposure to electricity up to that census year, and X_i is a vector of exogenous village characteristics, including years of exposure to primary and middle schools by 1996 and village topography.

As noted earlier, the OLS estimate of β may be biased if more developed villages – with lower fertility – were electrified earlier. For example, if a government policy prioritized extending electricity to more advanced villages, which tend to be less remote and have lower fertility, then E_i and the error term ε_i would be negatively correlated.

This would bias the OLS estimate of β downward, causing the effect of electrification on fertility to appear more negative than it truly is, or potentially to mask a positive effect.²

For the IV estimation, we first predict years of exposure to electricity as a function of elevation – which proxies for the cost of electrification – and other village characteristics. The first-stage regression is:

$$E_i = \alpha_0 + \alpha_1 Z_i + \alpha_2 X_i + v_i, \quad (2)$$

where Z_i is the elevation-based instrument and X_i is the vector of control variables described above.

The second-stage equation is:

$$Y_{it} = \alpha + \beta \hat{E}_i + X_i \psi + \varepsilon_i, \quad (3)$$

where Y_{it} is the village-level fertility outcome in year t (1996 or 2006), and $\hat{E}_i$ is the predicted years of electricity exposure from the first stage. The coefficient of interest, β , captures the causal impact of one additional year of electrification on fertility.

All regressions include district fixed effects, so the estimates are identified from variation across villages within the same district. This specification eliminates unobserved district-level characteristics – such as cultural, administrative, or policy factors – that may jointly influence both fertility and education.

5. Empirical results

5.1. OLS

The core of our empirical analysis compares OLS and IV estimates. We begin with the OLS results for the full sample, reported in Table 5. Recall that exposure to electricity is measured as the number of years since a village was first recorded as having electricity, up to 1996 and 2006. As expected, the OLS estimates of β in equation 1 are negative and statistically significant in both years. The results indicate that each additional year of exposure reduces fertility by about 3 children per 1,000 women in 1996 and 2 children per 1,000 in 2006. The smaller effect in 2006 reflects diminishing marginal gains from electrification over time. However, these effects represent only a small fraction of the observed fertility decline. Between 1986 and 1996, the average CWR fell by roughly 444 children per 1,000 women, meaning that even under the OLS estimates, electrification can explain only a minor part of the overall reduction in fertility.

The OLS estimates for village characteristics – including the availability of schools, TV, mosques, and religious composition – generally have plausible signs and magnitudes. Exposure to primary and middle schools is associated with lower fertility, and villages with TV signals also exhibit lower fertility in both 1996 and 2006. Shia-majority villages and those with a mosque report lower CWR as well, perhaps reflecting greater responsiveness to the information campaigns of a Shia-led government. Finally, villages in mountainous and forest areas have lower fertility than those in the plains, although the forest coefficients are not statistically significant.

²For a formal statement, see the online Appendix B.

Table 5. OLS estimates of the impact of electricity exposure on fertility using the full sample (dependent variable CWR)

	(1) CWR 1996	(2) CWR 2006
Electricity exposure	−0.003** (0.000)	−0.002** (0.000)
Primary school exposure (to 1996)	−0.002** (0.000)	−0.001** (0.000)
Middle school exposure (to 1996)	−0.001** (0.000)	0.000** (0.000)
TV exposure in 1996	−0.035** (0.004)	−0.025** (0.003)
<i>Village characteristics in 1986</i>		
Mosque	−0.033** (0.004)	−0.018** (0.003)
Shia majority	−0.052** (0.005)	−0.035** (0.003)
Log population	−0.013** (0.002)	−0.013** (0.001)
<i>Village geography</i>		
Mountain	−0.007* (0.003)	−0.016** (0.002)
Forest	−0.007 (0.008)	−0.000 (0.005)
Constant	0.792** (0.010)	0.507** (0.007)
Adjusted R^2	0.276	0.294
Observations	43,609	39,467

Notes: Exposure to electricity is measured to 1996 in column (1) and to 2006 in column (2). All columns use district-level fixed effects. Standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$.

5.2. IV

As noted in Section 3, the IV regression can only be estimated for the subset of villages for which elevation data are available. We therefore compare OLS and IV estimates using this smaller IV sample, reported in Table 6. In the OLS regressions, the coefficients on electricity exposure remain negative, and the estimates for other covariates are similar in both sign and magnitude to those obtained from the full sample.

A natural concern is whether the restriction to the IV sample introduces selectivity bias. However, as shown in Tables 2 and 3, the distributions of key village characteristics are very similar across the full and IV samples. We therefore conclude that the

when measured up to 2006. The corresponding F-statistics of 205.11 and 209.53 far exceed the conventional threshold of 10, ruling out concerns about weak instruments.

The key finding from the second stage is the positive coefficient on electricity exposure (Table 6). This result is consistent with the hypothesis that the small negative OLS estimates reflect downward bias; once corrected with the instrument, the effect reverses sign and becomes positive. In other words, rather than reducing the CWR by about 3 children per 1,000 women per year, electrification is estimated to increase the CWR by 22 per 1,000 in 1996 and by 16 per 1,000 in 2006 (columns 3 and 4). This reversal illustrates why IV estimation is necessary in this setting and underscores the importance of accounting for the non-random placement of electricity.

The IV estimates for most village characteristics – such as schooling and TV exposure, the presence of a mosque, and a Shia majority – retain the same signs as in the OLS regressions, although some coefficients are larger in magnitude. An exception is the mountain dummy: villages located in mountainous regions exhibit higher fertility, while those in forest areas do not differ significantly from villages in the plains. The coefficient on TV exposure is substantially larger in the IV specification, which may reflect the fact that TV access is highly correlated with omitted factors that also influence electricity exposure (the equivalent of δ in online Appendix B). Overall, the IV results confirm the robustness of the OLS patterns while underscoring the value of the IV approach in addressing omitted-variable bias.

6. Mechanism

The credibility of the hypothesis that electrification can increase fertility depends on demonstrating the mechanism through which this occurs and explaining why, in a specific context, the common empirical finding can be reversed. Economic theory suggests that a key channel through which infrastructure reduces fertility is by expanding market opportunities for women outside the home, thereby raising the opportunity cost of childbearing. In post-revolutionary Iran, however, such opportunities were contracting rather than expanding, which calls into question whether the impact of electrification in rural areas would follow the negative pattern observed elsewhere.

In principle, the absence of formal employment opportunities for women implies that electrification would not significantly increase the cost of childrearing. Under these conditions, the positive income effect of electrification could dominate, leading to higher fertility. Evidence from other settings supports this interpretation: for example, Lavy and Zablotsky (2015) show that when female LFP is very low, as among Arab women in Israel, the standard Beckerian mechanism linking education to lower fertility does not apply.

To test whether this mechanism accounts for the positive effect of electrification, we divide villages into low and high female LFP groups, depending on whether they are located in districts with female LFP above or below the median. District-level LFP is calculated from the 2% sample of the 1996 census, which reports a median rural female LFP of 8.6%. We then re-estimate equation 3 separately for these two sub-samples for CWRs in 2006. The choice of 2006 for the dependent variable is because the 1996 census is the earliest available in microdata, which makes the observation of CWR in 2006 more relevant.

The results, reported in Table 7, indicate that electrification increased fertility by 10 births per 1,000 women in high-LFP districts (column 2) and by 20 births per 1,000

lower education costs are a plausible consequence of electrification, increases in women's opportunity cost of time are highly context-dependent. In rural Iran, where female LFP was extremely low and restrictions on women's employment and mobility intensified after the Islamic Revolution, it is implausible that electrification raised the price of childbearing enough to offset the positive income effect.

In this paper, we provide empirical evidence that electrification in rural Iran during the 1980s and 1990s increased fertility, in contrast to findings from previous studies. Ordinary least squares (OLS) estimates suggest a negative effect, consistent with the literature, but correcting for the non-random placement of electricity using an instrumental variable (IV) approach reverses the result and reveals a significant positive effect. Our robustness checks – using clean water and clinics as proxies for child mortality, controlling for migration, and adding other infrastructure – consistently show that the positive effect of electrification on fertility is not an artifact of omitted channels but a robust causal relationship.

This positive causal relationship underscores that the demographic effects of infrastructure are strongly context-specific. Our results do not contradict prior research; rather, they clarify that electrification's impact depends critically on social structures, gender norms, and women's economic opportunities. Where women's participation in the labor market is severely constrained, the substitution effect of electrification (raising the opportunity cost of time) is minimal, allowing the positive income effect to dominate and increasing the demand for children.

From a policy perspective, our findings do not diminish the broader developmental benefits of electrification, nor do they deny that it can lower fertility through indirect channels. However, acknowledging the potential for a direct positive effect is essential when allocating scarce development resources. Policymakers should recognize that electrification alone may not reduce fertility, especially in contexts where complementary conditions – such as women's access to employment – are weak. In such settings, complementary policies that enhance women's economic opportunities, autonomy, and participation in the labor market may be necessary to achieve fertility-reducing outcomes.

In summary, while electrification remains an essential and highly beneficial development policy, its demographic impact varies substantially across contexts. Recognizing this complexity is crucial for designing effective development strategies. Future research could usefully explore the role of complementary policies – such as improving female education, expanding access to healthcare, and increasing women's labor market participation – in aligning infrastructure investments with desired demographic and developmental goals.

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What Is the Aggregate Economic Rate of Return to Foreign Aid?

Channing Arndt, Sam Jones, and Finn Tarp

In recent years, academic studies have been converging towards the view that foreign aid promotes aggregate economic growth. We employ a simulation approach to: (i) validate the coherence of empirical aid-growth studies published since 2008; and (ii) calculate plausible ranges for the rate of return to aid. Our results highlight the long run nature of aid-financed investments and the importance of channels other than accumulation of physical capital. We find the return to aid lies in ranges commonly accepted for public investments and there is little to justify the view that aid has had a significant pernicious effect on productivity. JEL codes: E1, O11, O41

Debate over the effectiveness of foreign aid as a tool to promote social and economic progress in developing countries has been sharp over many years. As noted early on by [Mosley \(1986\)](#), evaluations of foreign aid at the microeconomic and mesolevels have frequently found positive impacts. More recent studies in this vein support this view (e.g., [Michaelowa 2004](#); [Mishra and Newhouse 2009](#); [Arndt et al. 2015](#)). Controversies have been acute at the macroeconomic level. Even so, the large majority of up-to-date empirical studies in the economics literature have found positive impacts. More precisely, the full range of independent studies published since 2008 based on cross-country growth regressions report comparable results for the marginal effect of aid on growth. These studies

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suggest that receipt of foreign aid equal to 2.5 percent of GDP over a sustained period is expected to boost growth by approximately 0.25 percentage points on average. While some findings are statistically insignificant at conventional levels, in part reflecting noisy and sparse data, most are significant. Indeed, the broad magnitude and direction of these results are sufficiently similar to merit attention. This represents our point of departure.

In this study, we briefly review this new literature and go on to answer two specific questions: (i) are results from recent studies regarding the aggregate effect of aid on growth numerically coherent; and, if so (ii) what do they imply about the economic rate of return to aid? The first question is motivated by the notorious difficulty of pinning down causal effects in macroeconomic data. With respect to the assessment of foreign aid, these difficulties are compounded by the relatively low quality of data regarding income growth and foreign aid volumes in developing countries.

To answer the first question, we run numerical simulations of a dynamic neoclassical growth model, augmented with foreign aid. As deployed in various fields, from engineering (e.g., [Lin and Liu 1998](#)) to economics (e.g., [Ashraf et al. 2009](#); [Dalgaard and Erickson 2009](#)), numerical simulations offer a transparent means to determine plausible magnitudes of empirical phenomena. Also, numerical simulations can help to think through the empirical implications of specific modelling challenges, such as the suitable time frame over which aid impacts on growth. The second question focuses on the comparative benefits of providing aid versus the financial costs of its provision. A positive long-run impact of aid on growth does not automatically imply that aid generates an acceptable return on investment when viewed over its life cycle. At the same time, if returns to aid are found to be high, this might suggest that there is scope to provide a larger share of development finance on nonconcessionary terms. Even though rates of return represent a standard criterion for evaluating investments at the project level, these issues have not been addressed in the recent aid literature.¹ Our simulation approach can be used directly to calculate a plausible domain for economic rates of return to aid at the macroeconomic level.

The remainder of this article is structured as follows: Section I briefly reviews the set of peer-reviewed studies of the macroeconomic impact of aid on growth published since 2008. Section II presents our model, how it is calibrated, and the outcome indicators. Section III derives insights from selected simulations that focus on macroeconomic effects from physical capital investment, human capital upgrading, and productivity impacts associated with aid. Section IV considers the distribution of outcomes for both individual and combined aid impacts, based on Monte Carlo simulations. Section V summarizes the main findings and reflects on their implications. Two supplementary appendices (A and B), referred to in the text, are available online at <http://wber.oxfordjournals.org/>.

To preview our simulation results, we find that the marginal effect of aid on growth is negative over a nonnegligible share of observations when assessed over

1. See [Dalgaard and Hansen \(2005\)](#) for estimates of the marginal productivity of investment associated with aid.

a short time horizon (5 years). However, when the assessment window expands to 30 years, the macroeconomic effects of aid are consistently positive and highly comparable to findings from recent empirical studies. The average internal rate of return that corresponds to such results is around 11 percent. At the same time, we show that productivity and human capital accumulation effects are critical mechanisms through which aid can affect the macro-economy, especially over the long run. We conclude it is appropriate to view foreign aid as a long-term investment whose benefits cumulate slowly over long periods.

I. RECENT STUDIES

This section highlights the principal findings of recently published empirical studies that focus on the aid-growth relationship.² A meaningful starting point is Rajan and Subramanian (2008, hereafter RS08), who introduced a pair of influential innovations. First, they signaled a movement away from a reliance on cross-country dynamic panel data methods. Rather, their preferred strategy involved a long-run cross-section regression in which both Aid/GDP and growth are taken as averages over relatively extended periods (up to 40 years). This responds to the insight that aid given at time t may only have a growth impact after $t + n$ years, and that this impact may yield benefits over an extended period. Second, to address the endogeneity of aid, RS08 deploy external instrumental variables rather than the internal instruments commonly deployed in dynamic panel estimators.

Table 1 summarizes core results from recent papers that address aid and growth. To the best of our knowledge, the table covers the full population of studies that meet the following criteria: they (i) refer to an average aggregate aid-growth relation for developing countries as a group; (ii) include data spanning at least 30 years; (iii) attempt to address the endogeneity of aid; and (iv) are published in a peer-reviewed economics journal over the period from November 2008 to July 2014.³ As the various results included in the table use alternative specifications, an attempt has been made to select estimates from comparable models. In some instances nonlinear specifications involving a squared aid term are included. For these we report the marginal effect of Aid/GDP on growth evaluated by fixing Aid/GDP at 2.5 percent.⁴ In the final column, we report the probability associated with a two-tailed test that the reported point estimate (beta) is not different from zero.

2. More extensive overviews of the aid effectiveness literature can be found elsewhere. For example, see references in table 1, also Temple (2010); Arndt et al. (2010); Roodman (2007); Dalgaard et al. (2004).

3. It follows from these criteria that the coefficients reported in table 1 exclude: “naïve” OLS estimates, (even where reported in included studies); results that focus exclusively on selected regions, disaggregated aid measures or alternative outcomes; and estimates from (recent) working papers (such as Galiani et al. 2014).

4. This value is chosen as an approximation to the average annual value of aid received by developing countries over the last thirty years.

TABLE 1. Summary of Recent Aid-Growth Studies

Study ^a	Reference	Period	Spec. ^b	Beta	Std. Error	≈ Prob.
RS08	Table 4, col. 1	1960–2000	Linear	0.06	0.06	0.30
RS08	Table 4, col. 2	1970–2000	Linear	0.10	0.07	0.17
MR10	Table 4, col. 1	1960–2000	Linear	0.08	0.03	0.01
AJT10	Table 6, col. 2	1960–2000	Linear	0.09	0.04	0.02
AJT10	Table 4, col. 4	1970–2000	Linear	0.13	0.05	0.01
CRBB12	Table 7, col. 6	1970–2005	Nonlinear	0.15	0.06	0.01
CRBB12	Table 7, col. 10	1970–2005	Nonlinear	0.31	0.17	0.07
CRBB12	Table 9, col. 9	1971–2005	Nonlinear	0.27	0.13	0.04
CRBB12	Table 9, col. 9	1971–2005	Nonlinear	0.42	0.20	0.04
KSV12	Table 2, col. 5	1970–2000	Linear	0.05	0.05	0.32
LM12	Table 3, col. 4	1960–2001	Linear ^c	0.85	0.43	0.05
NDHKM12	Table 1, col. 4	1960–2006	Linear	−0.02	0.01	0.14
B13	Table 3, col. 1	1960–2000	Linear	0.12	0.04	0.00
B13	Table 3, col. 1	1970–2000	Linear	0.18	0.07	0.01
HM13	Table 2, row 1	1971–2003	Linear ^d	−0.01	0.00	0.00
AJT15	Table 1, col. 2	1970–2007	Linear ^e	0.30	0.18	0.09
Mean effect	Unweighted			0.19	0.06	0.00
	Weighted			0.12	0.04	0.00

^aRS08 is [Rajan and Subramanian \(2008\)](#); MR10 is [Minoiu and Reddy \(2010\)](#); AJT10 is [Arndt et al. \(2010\)](#); CRBB12 is [Clemens et al. \(2012\)](#); KSV12 is [Kalyvitis et al. \(2012\)](#); NDHKM12 is [Nowak-Lehmann et al. \(2012\)](#); LM12 is [Lessmann and Markwardt \(2012\)](#); B13 is [Brückner \(2013\)](#); HM13 is [Herzer and Morrissey \(2013\)](#); and AJT15 is [Arndt et al. \(2015\)](#).

^bFor nonlinear specifications (which involve a squared aid term) the marginal effect of Aid/GDP on growth is evaluated by fixing Aid/GDP at 2.5 percent; standard errors are also approximate for these cases.

^cFor comparability, this result is the marginal effect due to aid assuming no decentralization of government spending; however, the authors find that if the degree of decentralization exceeds 7 percent, the marginal effect of aid on growth is no longer statistically significant.

^dThis estimate controls for investment and is derived as an average from country-specific regressions.

^eBeta coefficient and standard errors are adjusted to raw values from standardized values as reported in the study.

Standard errors for the unweighted and weighted mean effects are derived from the set of beta coefficients reported in the table (see text). Probability is based on the normal distribution.

Source: Authors' collation from citations listed in Google Scholar.

In all but two cases, the beta coefficients in table 1 are positive.⁵ Three quarters are approximately significant at the 10 percent level, and just two are insignificant at the 20 percent level. The simple average of the point estimates for the average marginal effect of Aid/GDP on growth is 0.19; weighting by the logarithm of the inverse of their estimated variances yields an estimate of 0.12. Notably, these coefficients show substantial variation—their raw standard

5. The data underlying the first of these negative results (NDHKM12) has been reexamined and found wanting by [Lof et al. \(2015\)](#). The second negative result (HM13) derives from an estimation model that controls for aggregate investment, implying the estimated effect of aid on output is restricted to noninvestment channels.

deviation equals 0.21, which is suggestive of substantial (model) uncertainty. Nonetheless, following [Pesaran and Smith \(1995\)](#), one can use the standard deviation of a set of comparable regression coefficients to derive a nonparametric estimate of the standard error of their mean.⁶ Applying this method delivers an estimate of the standard error of the unweighted mean effect of 0.06. Correspondingly, the mean effect is statistically significant, which is consistent with the formal meta-analysis of 68 earlier aid growth studies found in [Mekasha and Tarp \(2013\)](#).⁷

Interestingly, the range of results reported in table 1 is consistent with the empirical estimates originally found in RS08, which sparked the recent wave of literature. Although these authors' own results are not statistically significant at standard levels, [Arndt et al. \(2010\)](#) correct the treatment of unreported values for aid flows from missing to null and show that plausible modifications to RS08's empirical strategy yield positive and significant estimates with the same data set.⁸ Furthermore, [Arndt et al. \(2015\)](#) find comparable and significant results using data updated to 2007 (table 1, final row). It also bears remarking that alternative econometric methods suggest similar conclusions. For example, drawing on country-specific (cointegrated) time series analysis for 36 African countries, [Juselius et al. \(2014\)](#) find that foreign aid had a statistically significant positive effect on investment or GDP (or both) in 27 cases. In an additional seven cases the effect on investment or GDP is positive but statistically insignificant; and in only two cases (Ghana and Tanzania) does the impact of aid appear to be negative and significant.⁹

Before proceeding, we note that regression-based estimates of the marginal effect of aid on growth can be taken directly as approximations of the internal rate of return (IRR) associated with aid.¹⁰ This relationship is set out formally in Appendix A (online supplement). While we take up calculation of the IRR in further detail in the next section, this approximation implies that the weighted mean effect in table 1 roughly translates to an IRR of 12 percent. In sum, therefore, recent empirical studies provide consistent support for the view that aid has had a positive average effect on growth (and economic return) when viewed over an extended time frame. The view that aid is ineffective finds much weaker support; and the notion that aggregate aid is actively harmful on average (e.g., see [Moyo 2009](#)) finds no endorsement in recent academic research.

6. This estimator assumes the coefficients (observations) are independent; thus the present results must be seen as approximate.

7. These authors find a positive and significant weighted average partial correlation between aid and growth, which is robust to adjustments addressing publication bias and moderator effects. Note that the estimates in [Mekasha and Tarp \(2013\)](#) are partial correlations (effect sizes) and thus are not directly comparable in magnitude to the estimates in table 1.

8. In fact, the correction from missing to null is sufficient to achieve statistical significance.

9. [Juselius et al. \(2013\)](#) review these two cases in further detail considering country-specific historical factors, leading to a different conclusion.

10. We thank an anonymous referee for suggesting this connection.

II. FRAMEWORK

Model

The previous section highlighted a notable degree of consistency across the full range of empirical studies published since 2008. An open question is whether these results are plausible—that is, given what we know about aid volumes, is the distribution of these estimates in accordance with reasonable assumptions about how aid affects growth? A preliminary answer is provided by RS08. Using a straightforward Solow-Swan framework, the authors derive an expression for the expected marginal effect of aid on growth, which is equal to the product of the share of aid invested, the share of capital in income and the output-capital ratio. Based on rough estimates of the latter for developing economies, RS08 conclude that a plausible range for the marginal effect of aid (as a share of GDP) on growth is from 0.08 to 0.16 but could be higher (lower) if aid enhances (undermines) productivity growth.

Following RS08 and other studies (e.g., [Dalgaard and Erickson 2009](#)), we retain a highly aggregate neoclassical growth framework and extend it in three main directions. First, we focus on the dynamic behaviour of the model rather than static results. This is relevant since, under the assumption that foreign aid contributes to growth through the accumulation of stocks of physical and human capital, marginal returns to these stocks may vary over time, as does the absolute value of associated depreciation charges. Second, there is a potential lag between receiving aid and its impact on growth, which is likely to depend on the proximate goals of aid. Indeed, we show that different assumptions about the duration of this lag have important implications for calculations of the effect of aid on growth and returns to aid, especially over short time horizons.

Third, we do not rely on a simplifying assumption that aid only affects growth through accumulation of physical capital. Although analytically convenient, research suggests that the process of transition of economies from low to high incomes simply cannot be understood purely in terms of accumulation of physical capital. Upgrading of economy-wide human capital is fundamental, particularly to shift production from lower to higher value added goods and services. Moreover, inclusion of this channel accords with the explicit interest of many foreign aid donors in social policy outcomes, such as education and health.

Our complete model is summarized as follows:

$$Y_t = \theta_t K_t^\alpha (h_t L_t)^{1-\alpha} \quad (1)$$

$$K_t = K_{t-1}(1 - \delta) + \delta K_0 + (1 - \gamma - \phi)A_{t-\mu} \quad (2)$$

$$A_t = \begin{cases} \lambda Y_t, & \text{if } 1 \leq t \leq t^* \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

$$c_t = (1 + \phi A_{t-1}/Y_{t-1})c_{t-1} \quad (4)$$

$$w_t = (c_{t-1} + 14w_{t-1})/15 \quad (5)$$

$$h_t = w_t^\eta \quad (6)$$

$$L_t = (c_t/c_{t-1})^\kappa L_{t-1} \quad (7)$$

$$\theta_t^* = 1 + \tau A_{t-1}/Y_{t-1} \quad (8)$$

$$\theta_t = \begin{cases} \theta_t^{*t/\omega}, & \text{if } t < \omega, \tau \neq 0 \\ \theta_t^*, & \text{otherwise} \end{cases} \quad (9)$$

$$K_0 = (r_0/\alpha)^{\frac{1}{\alpha-1}}; L_0 = 1; w_0 = 1; c_0 = 1; 0 \leq \gamma + \phi \leq 1$$

where t denotes time (in years; $t = 0$ is the initial or base year); Y_t denotes real national income; θ_t is total factor productivity (TFP); K_t is an aggregate measure of (physical) capital stock; L_t is a metric of labor quantity inputs (e.g., hours worked); h_t is an index of human capital quality; A_t is the volume of foreign aid; c_t is an index of the quality of human capital embodied in children; w_t is an index of workers' human capital; r_0 is the initial rental rate of physical capital. All human capital variables are normalized taking a base value of one.

With respect to time invariant parameters, $\alpha > 0$ is capital's share of income; $0 < \delta < 1$ is the rate of decay of physical capital; $\mu \geq 1$ is an integer that captures aid timing effects; $0 \leq \gamma \leq 1$ indicates the share of aid that does not contribute to the accumulation of domestic factors of production; $0 \leq \phi \leq 1$ gives the share of aid spent on human capital quality upgrading; $\lambda > 0$ is the share of aid in income (GDP); η reflects returns to worker human capital; $\kappa \geq 0$ indicates the impact of human capital expenditures on labor supply; τ is the terminal impact of aid on aggregate productivity; and ω denotes the period over which such productivity increases phase in (starting $t = 1$).

At the outset, five features of this framework can be highlighted. First, we assume there is no preexisting aid in the system ($A_0 = 0$), and the domestic saving rate is permanently fixed at δ . Furthermore, in the absence of aid, productivity and both the quality and quantity of human capital ($h_t L_t$) are constant. This means that the economy begins in a steady state and that foreign aid is the *only* lever able to shift the economy from its initial equilibrium. These assumptions are adopted not for realism; rather, they are a consequence of defining a simple model that focuses on the incremental contribution of aid.

It follows that a comparison of a given macroeconomic aggregate at $t > 0$ relative to its value at $t = 0$ will uniquely capture the total effect due to aid in a counterfactual sense.

Second, the model is framed as a closed economy that receives exogenous injections of funds in the form of foreign aid. We recognize this assumption is questionable. However, [Barro et al. \(1995\)](#) show that a credit-constrained neo-classical small open economy displays very similar dynamic properties to that of a closed economy. Thus, we believe critical insights are not lost by retaining a closed economy assumption. Also, we explicitly allow for human capital and productivity effects (through ϕ and τ , respectively), which are critical aggregate channels through which aid can influence growth in open economies, including via the real exchange rate. Third, capital's share of income remains constant over time, which is a direct corollary of Bowley's Law (e.g., [Krämer 2011](#)). Fourth, prices are presumed constant, normalized to one.

Fifth, a trade-off with adopting a simple model is that certain features are excluded. By definition, income (or other) shocks are ruled out, thereby ignoring the potentially important role of foreign aid in support of consumption smoothing. Also, we do not include disaggregated actors (e.g., firms, households, government) or their interactions. Certainly, such dynamics may be material in specific cases; however, they go beyond the present exercise which aims to establish broad orders of magnitude for aid's effects.¹¹ Moreover, despite the absence of utility maximizing agents, it remains appropriate to consider Solow-Swan growth models as dynamic general equilibrium frameworks (see [Acemoglu 2008](#)).

Looking in more detail, equations (1) and (2) are the basic Solow-Swan equations, expressed in aggregate (not per capita) terms and augmented by foreign aid. Equation (3) is an exogenous aid allocation rule. We treat aid as exogenous precisely because our aim is to investigate the counterfactual impact of aid on the macro-economy. Put differently, since we wish to verify the plausibility of recent econometric evidence regarding the causal impact of aid, it makes sense to undertake simulations "as if" aid were exogenous. Thus, the equation says that from $t = 1$ to t^* , the simulated economy receives aid inflows equal to a prespecified proportion of national income (λ). Terminating aid inflows at a specified period means that per capita growth will return to zero after some further time. In turn, this implies there is a finite horizon over which it is meaningful to study the aid-induced behaviour of the economy. As discussed below with reference to the calculation of returns to aid, this is analytically helpful since it delimits the relevant life-cycle of aid in the economy.

Equations (4) through (7) constitute the human capital block, in which the impact of aid operates through two main microchannels. At the intensive margin, we allow aid to enhance the quality of human capital embodied in

11. See [IMF \(2008\)](#) for examples of country-specific aid modeling exercises that include greater sectoral detail.

children, such as through education and nutrition interventions. As stated in equation (5), we assume one-fifteenth of the workforce is replaced in each period by the representative child, implying a long lag from aid delivery to its having a substantive effect on the quality of worker human capital that enters the production function.¹² At the extensive margin, we allow aid to increase hours worked, which can be thought of as incorporating some combination of an improvement in the disease environment and reductions in mortality. This effect is assumed to operate directly on existing workers and is cumulative. Moreover, we assume all human capital effects are permanent and thus permit the economy to reach a new steady state level of income after termination of aid flows. An implication of these two effects is that domestic savings (δK_0) will decline as a share of income—that is, we exclude an endogenous savings response. While this is conservative, our assumption of a zero rate of depreciation on human capital is more bullish. However, it accords with assumptions deployed by [Ashraf et al. \(2009\)](#) and avoids the complex question of how this should be parameterized (see [McFadden 2008](#)).

Finally, equations (8) and (9) represent a block of direct (aggregate) productivity effects. Specifically, equation (8) is a simple linear relationship that defines an aid-adjusted permanent level of TFP, which is achieved gradually as described by equation (9). These capture improvements in policies, institutions, or productive technologies, which are frequently the stated goals of donor funding. A related motivation for considering this channel is that, despite donor intentions, critics have often opined that aid may have negative consequences for aggregate productivity (e.g., [Moss et al. 2006](#); [Rajan and Subramanian 2007](#); [Djankov et al. 2008](#)). As such, this block can be considered an umbrella capturing a range of potentially important specific channels through which aid may affect productivity or competitiveness, which are not discussed explicitly here.¹³

Calibration

The above framework provides a basis to explore the most important generic mechanisms through which aid is often thought to have macroeconomic effects. To provide numerical insights, calibration of the model is necessary. In doing so, we stress that our objective is not to provide a single set of results that approximate a given country or circumstance. Our aim is to provide plausible bounds on the order of magnitude of the macroeconomic effects due to aid, considering individual mechanisms both separately and in conjunction.

A simulation of the model is defined by the values taken by the parameters; these determine the outcomes of interest. More formally, we define a parameter vector for simulation i as a draw from a joint distribution:

12. We recognize these demographic dynamics are crude. However, adding richer structure adds little analytical meat and therefore is excluded for parsimony.

13. Appreciation of the real exchange rate is one example. See [Arndt et al. \(2015\)](#) for further discussion and [Berg et al. \(2010\)](#) for a microfounded dynamic general equilibrium framework that includes some of these complex features.

$s_i \in \Omega = f\{t^*, \lambda, \alpha, \mu, \delta, \gamma, \phi, \kappa, \eta, \tau, \omega, r_0\}$. Evidently, Ω spans multiple dimensions. To reduce the dimensionality of the analytical problem, we keep parameters of secondary importance fixed throughout ($t^* = \omega = 30$; $\eta = 0.4$; see below). For the remaining parameters, insufficient prior information exists to completely specify their joint distribution. In particular, covariances and moments greater than two generally are unknown. Thus, we use existing literature to identify a plausible range of values. In most cases, we also presume an agnostic independent uniform distribution. This corresponds to Laplace's principle of insufficient reason and is intended to generate a broad prior over the parameter space.

Appendix table B1 summarizes the assumed ranges and distributions for parameters that vary between simulations. Capital's share of income ranges from 30 to 70 percent; the delay from aid delivery to it adding to physical capital is an integer uniformly distributed between one and seven periods (years). The share of income received in foreign aid is drawn from a truncated Beta distribution, taking a mean of approximately 5 percent, yielding a distribution with a large range that broadly reflects the varied pattern of historical aid flows. The duration of aid inflows (t^*) is held fixed at 30 periods, which is roughly an upper limit on the assessment window found in recent literature (see section I).

Choices regarding the rate of depreciation and marginal returns to capital are important and more controversial. Few rigorous estimates of depreciation rates are found in the literature, especially for developing economies. Bu (2006) uses data from enterprise surveys for a range of developing countries and finds aggregate capital stock depreciation rates of between 9 and 23 percent. These values compare to rates of 2 to 10 percent employed by corporate auditors in the same countries, typically also used in cross-country capital stock estimations (also Bu 2006). Faced with these ranges, we take an agnostic stance and assume depreciation rates are uniformly distributed between 2.5 and 25 percent.

For rates of return to physical capital, we rely on the calculations Caselli and Feyrer (2007). Their baseline estimates point to an average marginal product of capital, which is equivalent to the rental rate under the assumption of constant returns to scale and competitive capital markets, of 0.27 with a standard deviation of 0.09 among low-income countries. Notably, this estimate does not take into account "correction factors" such as differences in prices of capital goods and the absence of complementary factors. To some extent, these items can be considered as contributing to an effective depreciation rate. Here, the initial net rate of return $r_0 - \delta$ takes a mean of 13.2 percent and standard deviation of 11.0 percent, which indeed is much closer to the corrected rates of return estimated by the same authors.

The magnitude of health and education impacts on aggregate income, as well as the effects of foreign aid on these proximate drivers, remain contested. With respect to the latter, Arndt et al. (2015) estimate that sustained aid inflows of around 5 percent are associated with an increase in completed years of schooling among adults of between 2 and 3 years on average. Global estimates of schooling

provided by Barro and Lee (2013) indicate that improvements of this magnitude over a generation (30 years) are reasonable, especially for developing countries starting from a low educational base. For instance, average years of schooling (for the population 15+) in Botswana increased from 2.14 to 8.69 years from 1970 to 2000. Correspondingly, equation (4) indicates that if one half of a sustained aid inflow equal to 5 percent of GDP is spent on human capital upgrading ($\phi = 0.5$, $\lambda = 0.05$), child human capital quality would roughly double over a generation.

With respect to the economic impact of such investment, our model defines marginal returns to worker human capital quality as: $\partial Y_t / \partial w_t = \eta(1 - \alpha)Y_t / w_t$. Thus, as a proportion of gross income, an upper bound on these returns equals $\eta(1 - \alpha)$. For simplicity, to investigate the behaviour of the model we fix $\eta = 0.4$ throughout, implying a maximum per-period marginal return of 28 percent to increments in worker quality ($0.3 \leq \alpha \leq 0.7$). While this is larger than typical macro-Mincerian returns to education (see Lange and Topel 2006), it represents a strict upper bound and mechanically declines as w grows. Moreover, to the extent that upgrading of human capital generates economy-wide positive externalities, this calibration may prove conservative.

Aggregate income effects due to changes in morbidity or mortality are perhaps even more controversial. They are important, however, because demographic changes associated with large public health interventions can generate complex general equilibrium effects (see the discussion in Bleakley 2010). Ashraf et al. (2009) use burden of disease estimates (years lost to disability) to capture the effective loss of working time associated with poor health, ignoring premature mortality. Per capita figures compiled by the World Health Organization suggest this equals around 11 percent in low-income countries.¹⁴ Years lost to premature mortality are substantially larger, implying large health interventions could generate even more substantial labor supply effects over time. With this in mind, we set $\kappa \in [0, 0.15]$, implying a maximum labor force quantity increment of around 22 percent over a generation at the mean of simulated aid inflows.¹⁵

Lastly, the productivity parameter τ (equation [9]) is restricted to vary between zero and two. At the mean simulated value of aid inflows this implies a maximum terminal productivity increment equal to 10 percent of initial income, holding labor supply fixed. Since we assume this increment is phased-in over 30 periods, it equates to productivity growth of around 0.3 percentage points per year. Effects of this magnitude are not implausible. Long-run estimates of annual labor productivity growth in the US cluster around one percentage point on average (Nordhaus 2002). Moreover, in low-income countries, estimates of efficiency losses from factor misallocation are typically an order of magnitude larger (Vollrath 2009). That is, these estimates suggest large one-off gains might be achieved by suitable policy interventions.

14. See: http://www.who.int/healthinfo/global_burden_disease/estimates_regional/en/index1.html.

15. This is calculated using equation (7) as follows: $(1 + \phi\lambda)^{30\kappa} = (1 + 0.9 \times .05)^{4.5} = 1.219$.

For analytical purposes, we exclude negative values for τ , thereby imposing a positive correlation between productivity and aid inflows. We recognize this assumption is questionable; we adopt it because the rate of return calculations become highly unstable in the absence of positive cash flows. Nonetheless, because the first order effects of τ are symmetric around the origin (on average), our Monte Carlo simulation results can be used to consider the (average) magnitude of τ necessary for aid-induced productivity losses to fully dominate capital accumulation effects. In this light, it should also be pointed out that theoretical and empirical estimates of the impact of aid on competitiveness (e.g., via Dutch Disease) typically only point to weaker relative growth of manufacturing industries (Rajan and Subramanian 2011) and, at worst, stagnant aggregate income effects (Adam and Bevan 2006). As we show in sections III and IV, impacts of this kind are entirely consistent with negative values of τ within the mirror of the stipulated parameter range.

Outcomes

To analyze numerical outcomes from the model, two main indicators are in focus. The first is the marginal effect of aid on growth calculated over different periods. For run i and period length J , this is defined as:

$$g_{ij} = \frac{1}{\lambda J} \sum_{j=1}^J \dot{y}_{i,t+j} \quad (10)$$

where $\dot{y}_t = (Y_t/Y_{t-1})(L_{t-1}/L_t) - 1$, which can be thought of as a lower bound on the per capita growth income rate—that is, it is binding if effects due to equation (7) operate exclusively through improvements in mortality. This indicator corresponds to the principal focus of the empirical literature discussed in section I. In line with these studies we report results for 5, 10, and 30 periods. Thus, results for different window lengths provide insight as to how alternative econometric approaches to the problem may behave.¹⁶

The second indicator is the internal rate of return (IRR), which is calculated from the profile of domestic cash flows (net additional income) generated by aid. Under the maintained hypothesis that the economy is in a “no-aid” steady state at $t = 0$, the real value of the per-period cash flow attributable to aid is given by:

$$CF_t = Y_t - (Y_0 + A_t) = Y_t(1 - \lambda) - Y_0 \quad (11)$$

This says that the aggregate economic cash flow attributable to aid in a given year is the difference between aggregate income (with aid) at time t minus the

16. Panel estimators consider multiple consecutive windows across various countries. Even so, typically these rely on identification via first differences—that is, they focus on changes in growth relative to changes in aid inflows. Viewed in this way, only the initial period of the simulations relative to the baseline would be informative as this provides the primary variation in aid inflows.

sum of the counterfactual income (without aid) and the effective cost of aid in that period. This equation makes explicit our model assumption that all changes in the economy, including productivity and human capital effects, can only be induced by aid inflows.

From a project finance perspective, returns to investment depend on both the magnitude and time profile of such cash flows. For instance, although cash flows are expected to be negative in early years, viable investments should yield a positive net present value (NPV) of flows over the life cycle of the project. In other words, the key concern is whether the discounted sum of cash flows is positive.¹⁷ To calculate the NPV, the discount factor must be specified *ex ante*. If unknown, the IRR can be used to identify the discount factor which yields an NPV of zero. Intuitively, the IRR indicates the maximum cost of capital required for a project to break even. IRR calculations from simulations of our model thus indicate the maximum effective financing costs of foreign aid inflows that would generate no net losses. This interpretation suits our model as we assume all income gains associated with aid are available for consumption or payment of financing costs. For instance, if the IRR is found to be 10 percent, then the effective costs of financing aid over the lifetime of the project must be below 10 percent for such aid to have a positive overall net income effect.

Considering that estimates of the marginal effect of aid on growth represent approximations to the IRR (see section I; also Appendix A), one might query the added value of undertaking exact estimates of the IRR according to equation (11). In addition to the gain in precision, our motivation is that the approximation is only likely to be reasonable under highly restrictive conditions. To see this, note that the approximation assumes that the marginal aid-growth effect is constant and contributes both fully and immediately to growth in the first period aid flows arrive, which broadly corresponds to the basic model in RS08. It follows that once we allow aid to affect the economy through more complex (time-varying) channels, the quality of the approximation is likely to decline. Ultimately, however, this is an empirical question.

To complement the IRR, we also calculate two additional metrics. The first is the average ratio of net cash flow to initial income, denoted CFY. This is equivalent to the NPV calculated with a discount factor of zero, divided by the product of the initial income level and the number of periods in the simulation. Values greater than zero imply that the (undiscounted) cumulative cash flow from the project is positive—that is, the real value of income generated by aid is greater than the value of aid received, regardless of the time profile of positive and negative cash flows (to which donors may be indifferent). The second indicator

17. We define the project life-cycle as the period from the beginning of aid inflows ($t = 1$) to the period at which growth returns to a zero trend. For simulations where aid shifts the steady state income level, this termination rule ignores the ongoing (infinite) stream of benefits due to the changed equilibrium level of income. Note that for all simulations we impose a minimum life cycle of 60 periods, which can be thought of as comprising 30 years of aid inflows followed by a minimum of 30 additional years for impacts to ‘work through’ the economy.

depreciation rate of approximately 0.62 to nullify the 30 year aid-growth results, holding all other parameters at their simulation means.

V. CONCLUSION

This article took as its point of departure the substantial degree of consistency across the full range of recently published empirical studies that investigate the long-run marginal effect of aid on growth. Employing different methods and data sets, the large majority of these studies place this effect in the positive domain. The weighted average result from these studies indicates that a sustained inflow of foreign aid equivalent to 10 percent of GDP is expected to raise growth rates per capita by about one percentage point on average.

The first objective of this study was to investigate the coherence of these empirical results using numerical simulations. To do so, we proposed a dynamic general equilibrium model that captures a range of aggregate channels likely to influence the size and direction of the macroeconomic effects of aid. These included physical capital investment, consumption of aid, human capital upgrading—including quantity and quality mechanisms—and direct productivity effects. We then simulated the model to capture the effects of these channels, both individually and in combination. Selected simulation results were used to understand the behaviour of the model. These were augmented by Monte Carlo simulations, based on 5,000 random draws from the distributions of the core model parameters (constrained over reasonable ranges). These simulations were used to approximate the empirical distributions of key outcomes, principally the marginal impact of aid on growth and its internal rate of return (IRR).

When assessed over a short time horizon (5 years), the simulations revealed that the marginal effect of aid on growth is negative over a nonnegligible share of observations and has large variance. This reflects sensitivity to the lag structure through which aid effectively contributes to the economy. When assessed over the long-run (30 years), the macroeconomic effects of aid are consistently positive and distributed more tightly, especially from the point of view of the IRR. More specifically, both the mean and broad range of the simulated marginal effects of aid are highly comparable to findings from recent empirical studies. We also found that when aid is allowed to affect human capital quality, a time frame of a generation may not be sufficient to fully capture the economic contributions of aid.

The basic upshot of our analysis is that the overall picture given by the aid-growth empirics of table 1 is reasonable. It is compatible with a range of plausible parameterizations of a simple yet general model of the growth process. The central tendency of our simulations approaches the mean of recent empirical aid-growth studies, particularly when moderate positive effects from aid on human capital and/or aggregate productivity are permitted. In sum, both the latest empirical evidence and numerical simulations point to positive average growth returns to foreign aid when viewed over long time frames. The notion that aid

has broadly harmed development performance receives essentially no empirical support.

The IRR calculations associated with the same simulations require careful interpretation. On the one hand, these results might be considered fairly moderate. For instance, if only physical capital accumulation effects are permitted, as per our baseline model, the average IRR over 5,000 model runs is equal to 7.5 percent (table 3, row 2), while the corresponding 99 percent confidence interval ranges from -11.5 to 30.7 percent. Also, across all simulations the cumulative value of cash flows attributable to aid only turns positive after around 20 periods (at least). This reflects the lag structure of the model and that aid contributes only slowly to income via additions to stocks of capital. Simply put, long lags between the injection of funds and realization of economic benefits naturally militate against high IRRs.

From the perspective of both marginal growth effects and economic returns, these simulations underline the need for realistic expectations of foreign aid. Certainly, aid cannot be expected to deliver the kinds of high returns sometimes implied by two gap model exercises, which underpinned projections of rapid economic development in the early days of foreign assistance—a point vividly made by Easterly (1999; also Dalggaard and Erickson 2009). Similarly, some aid proponents have suggested that aid may yield very large economic returns when used to escape poverty traps. As Kraay and Raddatz (2007) note, however, these hopes appear not to have materialized. Observed reality, which is consistent with our simulation results, is that the journey from low- to middle-income status typically remains long and arduous, even when supported by effective aid inflows.

At the same time, predictions from poverty trap or basic two gap models are not required to provide an economic rationale for foreign aid. Our results imply that a sufficient criterion is its macroeconomic rate of return. The distribution of returns summarized in table 3 (also fig. 2), and which should be viewed as *ex post*, cluster around commonly used rates of return in *ex ante* project analysis such as the 10 percent cut-off applied by the World Bank (Pohl and Mihaljek 1992), or the seven percent cut-off employed by the United States Office of Management and Budget (Powers 2003). Recent growth empirics combined with a simple model of growth point to rates of return to aid that lie in a range commonly associated with successful long-run public sector investments.

Additionally, productivity and human capital accumulation effects appear to be important, if not fundamental, mechanisms through which aid can affect the macroeconomy. This is especially true when viewed over the long-run. For example, for moderate choices of the partial correlation between aid and the final level of TFP, the simulated distribution of the long-run marginal aid-growth effect more closely approximates the weighted average from recent empirical studies and the corresponding average IRR is significantly higher, at 13.1 percent. Allowing all parameters to vary simultaneously yields a mean IRR of 11.2 percent (table 3, row 1). Thus, whatever the combination of channels or their relative weight, it remains appropriate to view aid as a long-term investment